Cryptography Guessing Game

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Research Report

Submitted in partial fulfilment for the degree of
BSc (Hons) Computer Science

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8th April 2026

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Type of assessment	Individual
Coursework title	Project Proposal and Research Document (D3)
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ACKNOWLEDGEMENTS

I would like to thank my supervisor Dr. Mike Just for his guidance and feedback throughout the project.

ABSTRACT

This document presents the design, development, and evaluation of an educational web-based cryptography game, C1PHER, focused on classical ciphers. Classical cryptography is often taught using static explanations, which can fail to engage students. Gamification offers a more engaging and effective experience compared to conventional teaching and learning methods. Motivated by the lack of existing gamified tools for classical ciphers, this project aims to design and develop an educational web-based cryptography guessing game that challenges users to identify plaintext messages from ciphertexts encrypted using various classical ciphers. The game provides interactive gameplay, tiered difficulty levels, and a scoring system to help reinforce users' cryptographic knowledge.

The developed system incorporates key gamification elements such as progression through a structured journey, tiered difficulty levels, performance-based scoring, and immediate feedback to reinforce learning. Users engage with a range of classical ciphers through both informational content and challenges, supporting the development of cryptographic understanding in an engaging environment.

The project involved several key stages. First, an extensive literature review and background research were conducted to delve into the core concepts of a range of classical ciphers and the contrast to modern cryptography. It also explored the concept of gamification in education as well as existing games and tools for learning cryptography. Next, requirements analysis and system specification were carried out to ensure that the proposed game is feasible and meets the objectives. Following on, system design and interface prototyping were performed, informed by the games and tools researched, to help shape the user interface, gameplay mechanics, and overall user experience. The system was then implemented as a web-based application, followed by an evaluation through system testing and a user study to assess usability, engagement, and learning outcomes.

The results indicate that the integration of gamification and interactive challenges can enhance both engagement and perceived understanding of classical cryptography. Overall, this work demonstrates the potential of gamified learning approaches in making technical subjects such as cryptography more accessible and engaging.

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1 Introduction

1.1 Background and Motivation

Cryptography is the practice of developing and implementing algorithms to secure and conceal information, ensuring that it can only be read by those with the permission and ability to decrypt it [1]. Technology is prevalent in today's society, and with so much of our daily lives being online, it is essential that people can learn and understand how their information is protected. The loss of unprotected data can be devastating for individuals and businesses alike.

The cryptosystems used today are complex and confusing. It doesn't make sense to try to understand an algorithm like AES without understanding the first algorithms that inspired it, such as the shift cipher. Traditional learning resources, such as textbooks and online tutorials, often fail to capture learners' interests or inspire them. The motivation for the project stems from this. Cryptography is an interesting and exciting field, but it doesn't take long for concepts to become confusing and frustrating to learn. By using gamification, learning about cryptography can be made "more attractive, captivating and, ultimately, effective" [2].

1.2 Aim and Objectives

This project aims to design and develop an educational web-based cryptography guessing game that challenges users to identify plaintext messages from ciphertexts encrypted using various classical ciphers. The game will develop and test users' cryptographic knowledge through interactive gameplay, tiered difficulty levels, and an integrated scoring system.

To fully achieve this aim, the following objectives have been established:

O1: Design and implement an interactive web-based interface that enables users to view cipher information, attempt to decrypt ciphertexts and receive real-time feedback on their guesses.

O2: Incorporate multiple classical encryption algorithms, each accompanied by educational content, to generate diverse cryptographic challenges.

O3: Develop a structured difficulty system that categorises challenges by cipher and complexity, providing optional hints to support users' learning and problem-solving.

O4: Implement a points and leaderboard system that awards users based on the difficulty of completed challenges and displays global rankings to promote engagement and competition.

O5: Design a clear, intuitive and responsive user interface that ensures accessibility and usability across a range of major web browsers, including Google Chrome, Mozilla Firefox, Apple Safari, and Microsoft Edge.

O6: Develop a secure backend system capable of managing user accounts, storing progress, scores and ranking data in an external database.

O7: Conduct functional testing and pre- and post-development usability evaluations to ensure that the game is reliable and effective as a learning tool.

1.3 Organisation

The remainder of this dissertation is structured as follows:

- Section 2 – Background Research: This section explores the theoretical and contextual foundations relevant to the project. It introduces key concepts in cryptography, reviews classical and modern ciphers, explores the concept of gamification in education and examines existing educational tools and games.
- Section 3 – Work Plan: This section outlines the research goals, functional and non-functional requirements, and planned tasks for the project. It also presents the evaluation strategy used to assess the usability, engagement, and educational effectiveness of the developed system.
- Section 4 – Feasibility: This section discusses the technical feasibility of the project, detailing the chosen development environment, programming languages, frameworks, and database system. It also describes the preliminary prototypes and design considerations that informed the implementation process.
- Section 5 – Implementation: This section details the development of the system, including its software architecture, core components, and key design decisions as well as how challenges were addressed during development.
- Section 6 – Evaluation: This section presents the evaluation of the system, including requirements validation, system testing, and results from the user study. It analyses the findings in relation to the research questions and provides a critical discussion of the system's effectiveness.
- Section 7 – Conclusion: This section summarises the overall project, highlighting key contributions, findings, and limitations. It also reflects on the projects outcomes and identifies areas for future work.

2 Background Research

This section will explore theoretical and contextual concepts essential for developing the cryptography guessing game. Section 2.1 provides an overview of cryptography, introducing key concepts and terminology that are essential for understanding encryption and decryption processes. Section 2.2 then examines classical ciphers, exploring traditional techniques such as the Caesar and Vigenère ciphers that form the basis of the challenges that will be implemented in the game. Following this, Section 2.3 discusses modern cryptography, outlining how ciphers have evolved in the modern era. Section 2.4 reviews educational tools and games relevant to cryptography, identifying gaps that this project aims to address. Finally, Section 2.5 concludes the chapter with a summary of the key insights that inform the design and methodology of the proposed cryptography guessing game.

2.1 Cryptography Overview

Cryptography is the study and practice of secure communication over an insecure channel using mathematical techniques to transform readable data into an unreadable form. "An encryption algorithm, or cipher, is a means of transforming plaintext into

ciphertext under the control of a secret key. This process is called encryption or encipherment. Encryption can be expressed using the formula $c = e_k(m)$, where m represents plaintext, e is the cipher function, k is the secret key, and c is the ciphertext. The reverse process, known as decryption or decipherment, is simply $m = d_k(c)$" [3]. Generally, the algorithms for encryption and decryption are public knowledge, and it is the secrecy of the key that protects the plaintext.

Cryptographic systems can be categorised into two types based on the number of keys they utilise. Symmetric cryptosystems use a single key, which must be known to both parties encrypting and decrypting data. To contrast this, asymmetric cryptosystems use two different keys. The key used for encryption is publicly available, and the key used for decryption is private.

Understanding the distribution of English letter frequencies, as well as second-order statistics about the underlying language, such as the most common sequences of two or three letters, known as bigrams and trigrams, is crucial to cryptanalysis (Appendix D.1, D.2, D.3). Recognising predictable linguistic patterns can aid in deciphering monoalphabetic substitution ciphers [3].

2.2 Classical Ciphers

Classical cryptography, used before the development of computers, can be broken down into the following categories: monoalphabetic substitution ciphers, transposition ciphers, polyalphabetic substitution ciphers, described below, as well as fractional ciphers, and digraph substitution ciphers.

2.2.1 Monoalphabetic Substitution Ciphers

In monoalphabetic substitution ciphers, each letter of the plaintext is replaced by a fixed substitute. Examples include the Atbash, Pigpen, Shift (Caesar), Affine, and Mixed Alphabet Ciphers.

The Shift Cipher is one of the earliest forms of encryption, where each letter in the alphabet is shifted by a fixed number of positions. When the shift value is three, it is often referred to as the Caesar Cipher [4]. The method involves assigning the letter A the number 0, the letter B the number 1, and so on, until Z is assigned the number 25. The plaintext message is then converted into a sequence of numbers. To acquire the ciphertext, the secret key k modulo 26 is added to each number, and the resulting sequence of numbers is then translated back into letters. Since there are only 26 possible keys, the shift cipher can be easily broken through brute force [5].

The Substitution Cipher generalises this idea by allowing any permutation of the alphabet as the key. This results in 26! possible keys, the large key space is far too vast to allow for a brute-force search, even with a modern computer. However, the substitution cipher remains vulnerable to letter frequency analysis [3].

The Pigpen Cipher, famously associated with the Freemasons in the 18th century, replaces letters with geometric symbols. It has also been linked to Union prisoners during the American Civil War. It has a presence in popular media, featuring in books, games, and shows, such as the game Assassin's Creed II and the BBC's Sherlock. The Atbash

Cipher, which was one of the earliest recorded substitution ciphers, was used for the Hebrew alphabet. To encrypt the plaintext, the alphabet was reversed [6].
The Affine Cipher applies a more mathematical approach. Each of the letters in the plaintext is transformed to its corresponding integer in the range 0 to $m - 1$. Then, each letter is encrypted using the function $E(x) = (ax + b) \bmod m$, where a and b are the keys for the cipher. Interestingly, not all values of a and b work [7].
The Mixed Alphabet Cipher contrasts these ciphers by using a keyword or key phrase instead of a number as a key. Encryption works by first creating a random order for the ciphertext alphabet. The ciphertext alphabet is generated using the keyword/key phrase as follows: the keyword is first written, ignoring any repeated letters, and then the remaining letters of the alphabet are written in alphabetical order. Keeping this in mind, the chosen keyword should contain letters spread across the entire alphabet [8].

2.2.2Transposition Ciphers

Unlike substitution ciphers, a transposition cipher rearranges the position of letters without altering the actual characters. One of the oldest examples is the Spartan Scytale, the first-ever military cryptographic device used by the Ancient Greeks. The scytale is a wooden staff around which a strip of leather or parchment is wound, only when wound around the staff would the message make sense [9]. Other examples include the Rail Fence, Route, Columnar Transposition, and Myszkowski Transposition Ciphers. The Rail Fence Cipher writes the plaintext in a zigzag pattern, then reads it row by row [10]. The Route Cipher arranges letters in a grid and reads them in a predefined pattern. The Union Route Cipher, used by the Union forces during the American Civil War, is a variation of this. Instead of transposing letters by the given route, whole words were moved around, sometimes adding whole null words to the ciphertext to help obscure meaning.
The Columnar Transposition Cipher involves writing the plaintext out in rows and then reading the ciphertext off in columns. In its simplest form, it is the Route Cipher, where the route is to read down each column in order. However, Columnar Transposition usually uses a keyword for encryption. The method for this is writing the plaintext in a grid where the number of columns is equal to the number of letters in the keyword. The columns are then read in alphabetical order of the keyword. Sometimes nulls are used to enhance the grid [11].
The Myszkowski Transposition Cipher is a variation of the Columnar Transposition Cipher. The only difference is that when the keyword has repeated letters, the same number is assigned to them. These identical letters are then read in sequence rather than from left to right.

2.2.3Polyalphabetic Substitution Ciphers

To overcome the weaknesses of monoalphabetic substitution, polyalphabetic ciphers use multiple substitution alphabets. An example of this is the Vigenère Cipher. This cipher uses a keyword to determine a sequence of shifts applied repeatedly across the plaintext. The keyword is repeated to match the message length, forming a keystream. The

ciphertext is formed by shifting each letter of the plaintext according to the corresponding letter of the key.
Kasiski Analysis is a method of finding the length of the keyword and is therefore a method for decrypting the Vigenère Cipher. It works by examining distances between repeated sequences, which should be a multiple of the keyword. Taking the greatest common divisor of these distances can reveal the length of the keyword, allowing then for frequency analysis to be applied [12].
The Vigenère Cipher uses a tool called the Vigenère Square (also known as the Tabula Recta). The square is a 26x26 grid of shifted alphabets that provides the multiple substitution alphabets used in encryption (Appendix D.4). The keyword dictates which row is selected for each character of the plaintext. This means there are varying shifts across the plaintext, hence why the Vigenère Cipher is a polyalphabetic substitution.

2.3 Modern Cryptography

Classical ciphers are not secure enough to protect data in the modern era; however, they remain the essential building blocks for the much more complex algorithms that are designed to meet the security demands of today's digital world. To be able to understand the ciphers used today, an understanding of classical ciphers is required. These same ideas of substitution and transposition reappear in modern systems, but on a much larger and more mathematical scale, giving rise to symmetric block ciphers. Some examples of these are the Advanced Encryption Standard and Twofish, which provide robust protection for data by transforming it through multiple rounds of mathematical operations and key-based transformations.

2.3.1 Advanced Encryption Standard (AES)

AES (Advanced Encryption Standard) is a symmetric encryption algorithm widely used today for securing internet communication, file encryption, VPNs, wireless security, and data storage due to its balance of efficiency and robustness [14].

2.3.2 Twofish

Twofish is another symmetric block cipher. It was popularized by the algorithm DES and is the basis for a lot of block ciphers published since then [16]. Twofish is not as widely used as AES due to its slower performance. However, it remains in use for applications such as OpenPGP email encryption, VeraCrypt file encryption, and some VPN protocols [17].
While modern algorithms such as AES and Twofish demonstrate how classical techniques scale into complex systems, they are not directly relevant to the educational focus of this project.

2.4 Gamification and Educational Tools for Cryptography

Gamification refers to the use of game design elements in non-game contexts to enhance user engagement and motivation. Within educational settings, it has been identified as a promising approach for increasing student engagement and improving learning outcomes. By incorporating elements such as rewards, progression systems, and

structured challenges, gamified systems encourage users to actively engage with content rather than passively consume information [18].
However, the effectiveness of gamification depends on how well these elements are aligned with educational objectives. Simply introducing rewards is not sufficient; meaningful feedback and appropriately structured challenges are necessary to support understanding and skill development.
To make cryptography more engaging and accessible, several educational platforms and games have been developed. Some examples of these are as follows: CryptoHack, TheCODEBREAKERS, Cypher, Ink Cipher, Prose and Codes, and Kryptic.
CryptoHack offers a series of interactive puzzles and challenges that aid in teaching modern cryptography concepts through hands-on problem solving. Players learn about vulnerabilities and implementation details of real-world cryptosystems by breaking progressively complex ciphers [19].
TheCODEBREAKERS is an international online cryptography game dedicated to the conquerors of the Enigma Cipher, M. Rejewski, J. Rozycki, H. Zygalski and other members of the Cipher Bureau team. It introduces players to a range of historical ciphers, allowing them to experiment with different encryption and decryption techniques while following a narrative inspired by real events [20].
Cypher is a first-person puzzle game that takes place in a virtual museum dedicated to cryptography. Players progress through exhibits that teach the history and logic of different cipher systems, from simple substitution ciphers to the Enigma machine, combining exploration with education [21].
Other projects, such as Ink Cipher, Prose and Codes, and Kryptic, blend entertainment and learning in unique ways, whether through story-driven puzzles, world-based challenges, or web-based cipher-solving activities. However some of these tools are not fully accessible to users with little or no prior knowledge of cryptography, with steep difficulty curves making later challenges particularly demanding. For example, Cypher offers limited guidance and relies on users to independently research solutions, which reduces accessibility for beginners. Similarly, while CryptoHack is an effective gamified learning platform, it focuses primarily on modern cryptography rather than foundational classical ciphers. TheCODEBREAKERS also provides a strong educational experience, but its collaborative elements may not appeal to all users.
While these examples demonstrate the potential of gamified approaches to teaching cryptography, they also highlight a gap in tools that provide a structured, accessible introduction to classical ciphers. These limitations directly inform the design of the proposed system, which aims to balance accessibility, guided learning, and progressive challenge to better support users with varying levels of prior knowledge.

2.5Summary of Background Research

This background research explored the evolution of cryptography from its classical origins to modern implementations. Classical ciphers, including substitution and transposition methods such as the Shift, Substitution, Pigpen, Atbash, Affine, and Mixed Alphabet ciphers, established foundational techniques in data security and cryptography. These systems demonstrated key principles such as frequency analysis and linguistic

pattern recognition, which underpin modern cryptographic thinking. The transition to modern cryptography introduced mathematically complex algorithms, such as AES and Twofish, providing secure and efficient data protection for digital communications.
In parallel, the use of gamification and educational tools has emerged as an engaging approach to teaching cryptography. Platforms such as CryptoHack and TheCODEBREAKERS combine interactive problem-solving with historical and conceptual learning. However, while these tools enhance accessibility and engagement, many emphasise advanced or puzzle-oriented challenges rather than structured learning of foundational ciphers. This highlights an opportunity for a gamified system, such as the proposed system, that focuses explicitly on classical cryptography and introduces concepts through an incremental, feedback-driven learning experience. These findings highlight the potential of combining classical cryptography with gamified learning approaches. In particular, elements such as progression, feedback, and interactive challenges inform the design of the proposed system, which aims to deliver an engaging and educational user experience.

3 Work Plan

Building on the background research, this section outlines the design approach used to translate these concepts into a functional educational system. In Section 3.1 the research goals are outlined and discussed. Then, in Section 3.2 the functional and non-functional requirements are listed and prioritised. Section 3.3 discusses the tasks required to achieve the projects objectives and address the research questions. Following that is Section 3.4, the evaluation strategy. Finally, Section 3.5 summarises the chapter.

3.1 Research Goals

The purpose of this research is to explore how classical cryptography methods can be effectively taught and reinforced through an interactive, gamified learning environment. To guide this investigation, the following research questions have been formulated:
RQ1: How can gamification make the learning of classical cryptography more engaging and accessible to users?
RQ2: What combination of cipher complexity and instructional guidance optimally supports users' learning and problem-solving performance?
RQ3: Which cipher types and difficulty levels provide the most effective balance between educational value and playability?
RQ4: What feedback mechanisms and scoring systems effectively reinforce users' cryptographic understanding and influence user motivation?
RQ5: To what extent does the developed game improve users' understanding of classical ciphers compared to traditional static learning materials?
Answering these questions will help establish how classical cryptography concepts can be presented in an interactive and motivating way. The findings will contribute to designing an educational tool that utilises gamification to teach cryptography, while also providing insight into how these approaches compare to conventional instructional methods.

3.2 Requirements

The functional requirements focus on core gameplay and learning interaction, while non-functional requirements address system performance, usability, and scalability. Together, these ensure that the system is both educationally effective and technically viable.

Each requirement has been assigned a priority. High priority (H) is something that must be done; medium priority (M) is something that would be nice to have but is not essential; low priority (L) is something that could be done but will not impact the core functionality of the system. Each requirement is directly related to achieving the project's primary objectives specified in Section 1.2.

Table 1: Functional Requirements

ID	Description	Priority
FR1	The system must display the information of the selected cipher. (O1)	H
FR2	The system must allow users to access a challenge where they attempt to decrypt ciphertexts. (O1)	H
FR3	The system must provide real-time feedback on the user's guesses. (O1)	H
FR4	The system must include multiple classical encryption algorithms to generate diverse cryptographic challenges. (O2)	H
FR5	The system should include optional hints to assist in solving the cryptographic challenges. (O2)	M
FR6	The system must categorise challenges by cipher type and difficulty level where applicable. (O3)	H
FR7	The system must award points based on the difficulty of the completed challenge. (O4)	H
FR8	The system could maintain a global leaderboard that displays users' total points. (O4)	L
FR9	The system could allow users to register, log in, and log out securely, enabling them to track their progress and scores. (O6)	L
FR10	The system could store users' accounts, scores, and progress data in an external database. (O6)	L
FR11	The system should support multiple concurrent users without data loss or interference. (O6)	M
FR12	The system should allow users to view their progress history, including previously completed challenges and scores. (O6)	M
FR13	The system could provide an admin interface for managing challenges, ciphers, and educational content. (O6)	L
FR14	The system must provide clear and accurate error messages and feedback when invalid inputs are entered or when decryption attempts fail. (O1, O5)	H

Table 2: Non-Functional Requirements

ID	Description	Priority
NFR1	The system must comply with GDPR. (O6)	H
NFR2	Users' data must be stored in a secure database with restricted access, minimising vulnerability to security threats. (O6)	H
NFR3	The system's features must be reliable, ensuring that the game remains available and functional most of the time, only being unusable when updates are necessary. (O6, O7)	H
NFR4	The system must back up user data daily at midnight to ensure it is recoverable in the event of a crash. (O6)	H
NFR5	The system should be scalable to handle thousands of users and devices concurrently. (O6)	M
NFR6	The system must respond to user inputs and requests within one second or less under normal load conditions. (O1, O7)	H
NFR7	The system should be responsive in most common browsers, such as Google Chrome, Mozilla Firefox, Apple Safari, and Microsoft Edge. (O5, O7)	M
NFR8	The system should be responsive on mobile and tablet devices, such as Android, iOS and Windows. (O5, O7)	M
NFR9	The system must provide clear and user-friendly error messages to guide users in cases of invalid inputs or failed operations. (O1)	H

These requirements are directly informed by the research findings and aim to balance educational value while engaging gameplay mechanics.

3.3 Tasks

To achieve the project's objectives specified in Section 1.2 and address the research questions outlined in Section 3.1, the following tasks have been planned. Each task represents a distinct phase of work and contributes to the successful design, development, and evaluation of the proposed cryptography guessing game.

T1: Literature Review and Background Research - This initial stage of the project involved conducting background research on classical and modern cryptography, as well as reviewing existing gamified learning tools.

T2: Requirements Analysis and System Specification - The requirements analysis and system specification stage involved defining and refining the system's functional and non-functional requirements. These outline the features of the cryptography game.

T3: System Design and Interface Prototyping - Having completed task 2, the next step is to design the overall system architecture and user interface. The prototyping will explore layouts, and interaction designs to promote usability, accessibility, and engagement.

T4: Implementation and Integration - This stage involves developing both the frontend and the backend.

T5: Testing and Evaluation - After implementation, testing will be conducted to ensure the system functions as intended and all the features work correctly. Additionally, usability testing will be undertaken to ensure the game is both accessible and engaging.
T6: Reflection and Documentation - The final phase of the project involves analysing the collected data, interpreting the results in relation to the research questions, and producing comprehensive documentation.

3.4Evaluation Strategy

The evaluation of this project will assess the usability and educational effectiveness of the game. It will focus on how effectively the system supports users in learning about classical ciphers, while ensuring that the interface is intuitive and engaging. The evaluation will comprise of two stages: a pre-development evaluation (where feasible) and a post-development evaluation, as well as a technical evaluation of the final system.

3.4.1Pre-development Evaluation (Prototype Testing)

Before development begins, preliminary user interface mock-ups and/or interactive prototypes will be created to explore potential gameplay concepts and layout designs. Participants will be invited to review these mock-ups and provide feedback through online questionnaires. Feedback from this stage will be used to refine and improve the design concepts before they are implemented. However, this stage may be omitted due to time constraints.

3.4.2Post-development Evaluation (System Testing and User Study)

Once the game has been developed, a structured user evaluation will be conducted to assess the system's usability, engagement, and learning impact. Participants will be asked to complete specific tasks within the game, such as decrypting ciphertexts using the provided information and hints as needed. These tasks may be timed, and then participants will be asked to complete questionnaires evaluating the interface, accessibility, and overall satisfaction. This stage will provide both quantitative data (e.g., task completion time, rating scores) and qualitative feedback (e.g., open-ended responses), enabling a comprehensive assessment of the system's performance and educational effectiveness.

3.4.3Technical Evaluation

In addition to user-based testing, the developed system will undergo technical evaluation to ensure its functionality and stability. This will include:

- Functional testing to verify that all features (e.g., login, level progression, score tracking) operate correctly.
- Performance testing to confirm smooth operation and responsiveness across common web browsers and devices.
- Accessibility testing to verify that the interface supports different screen sizes and adheres to accessibility best practices.

3.5 Summary of the Work Plan

This chapter outlined the plan for this project, including the main tasks to be undertaken and the evaluation strategy. In the next section, the project's feasibility will be demonstrated.

4 Feasibility

This section covers the feasibility of the project. In Section 4.1, the coding environment is outlined. Section 4.2 discusses the design and database architecture. Following on from this, Section 4.3 presents the prototypes that have been developed. Finally, Section 4.4 summarises the chapter.

4.1 Coding Environment

The project will be implemented as a web-based application using the following programming languages:

- HTML
- CSS
- JavaScript

Development will be conducted using Visual Studio Code (VS Code).
Regarding libraries, frameworks, and/or packages, none have been intended for use currently, however, this may change as development progresses.
The primary focus of the project is to develop a functioning frontend, however, if time allows, a backend will be implemented to enable users to create accounts and view their progress on a global leaderboard. This will most likely be done using Node.js.
The system will use MariaDB as the database management system.
Game assets will primarily include:

- Text-based cipher challenges.
- Minimal visual elements (e.g., icons, backgrounds, and interface buttons).

All assets used will be appropriately licensed and attributed where necessary.
These components have been chosen due to familiarity with most of them, aiming to save time during the development phase by avoiding extended periods of learning a new software or language.

4.2 Design

The following subsections will outline the planned design of the proposed system, including the software architecture and the database structure. The design aims to ensure that the project is technically achievable, maintainable, and user-friendly.

4.2.1 Software Architecture

The system follows a client-server architecture, separating the game's logic, presentation, and data storage layers.

- The frontend, developed with HTML, CSS, and JavaScript, provides the user interface where players interact with cipher challenges and view feedback.

- The backend (to be implemented if feasible) handles user authentication, and leaderboard data.
- The database stores cipher data, user information, and score records.

4.2.2Database Design

The database (Appendix D.5) is designed to efficiently store user and game-related data while supporting scalability for multiple concurrent users. The key entities are as follows:

- Users – stores the user ID, username, email, password (hashed), percent completion, and total score.
- Ciphers – stores cipher ID, and cipher name.
- Challenges – stores the challenge ID, the cipher ID, the challenge answer and the score for completion.
- User Progress – stores user ID, challenge ID, the time of completion and the score awarded.

4.2.3Summary

The proposed design supports modularity, allowing for each feature to be developed and tested independently. This reduces complexity and allows for additional ciphers or features to be added in later stages. The architecture is intentionally lightweight to ensure efficient development during the allotted timeframe. Feasibility is further supported by the use of familiar technologies (HTML, CSS, JavaScript, GitHub, and MariaDB), which minimises setup and learning time, also, optional backend components can also be deferred if necessary, helping to maintain a manageable project scope.

4.3Prototypes

This section presents early prototypes developed to validate design decisions and test technical feasibility. The prototypes were built using Figma. The prototypes were developed to:

- Test the layout and usability of the web interface.
- Explore visual and interactive elements before backend integration.

Below are a few of the planned screens to be included in the web game, including the Cipher Journey screen, an example of a cipher's explanation page and a challenge page.

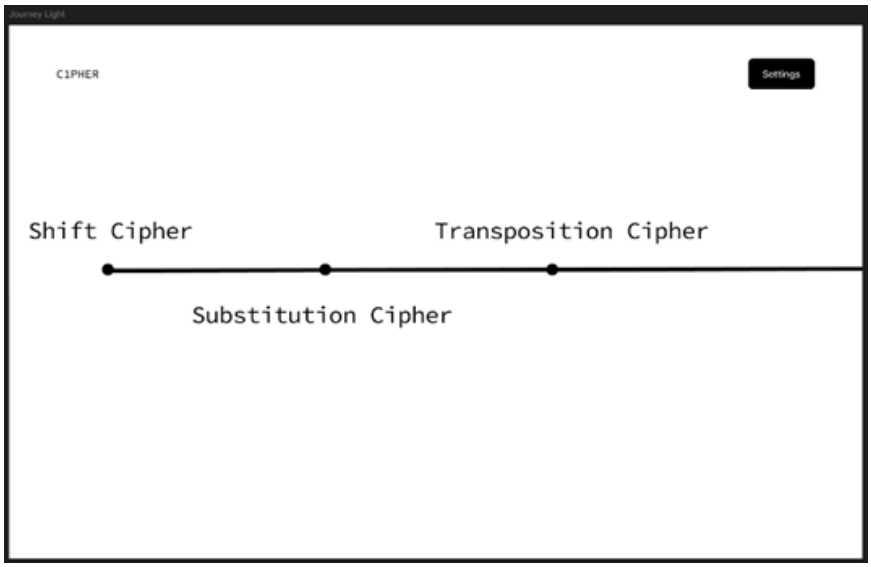

Figure 1: Journey Page (Light)

The screens included show an early idea of how the layout will look. The journey page (Figure 1) will include a horizontal 'timeline' that players will work 'chronologically' along to complete their journey in learning classical ciphers. By clicking on the cipher name the users will navigate to the chosen cipher's explanation page. This will include information about the cipher needed to understand how to encrypt and decrypt text. An example of an information page can be seen in Figure 2. This page will also include buttons that allow users to navigate to the challenge or challenges for that cipher (Figure 3). Completing these challenges will show on the Journey page and award the players points that accumulate for a total score. This total score will be what places users on the planned leaderboard. The login/ sign up page can be found in the appendix as well as the dark mode of the web pages (D.6).

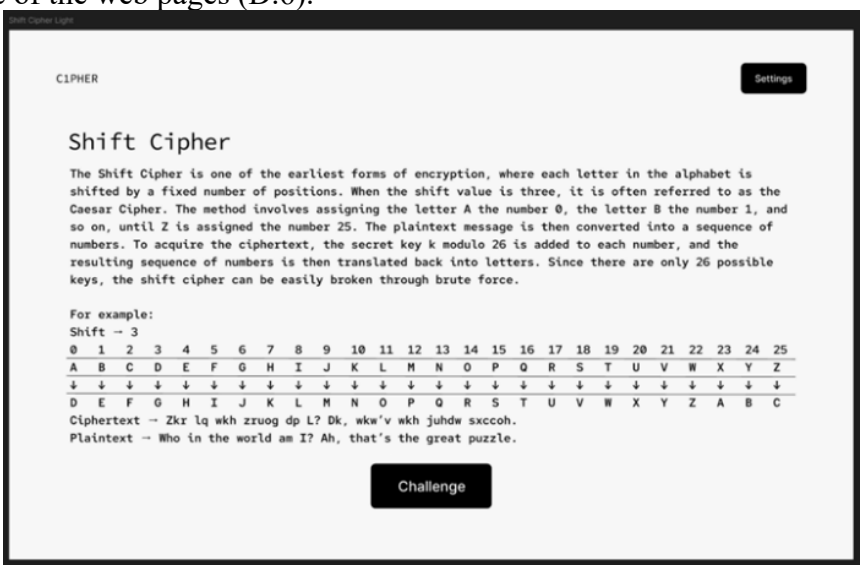

Figure 2: Explanation Page (Light)

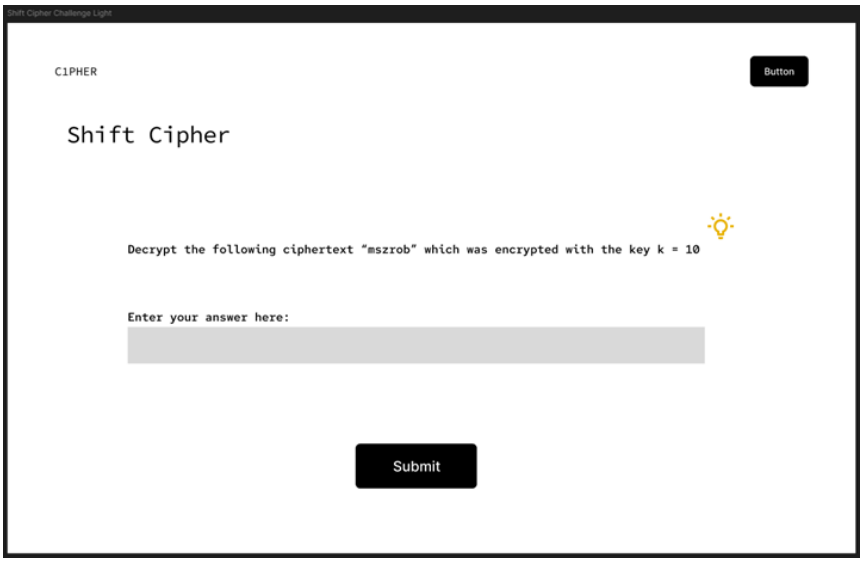

Figure 3: Challenge Page (Light)

4.4Summary of Feasibility

This chapter has outlined the technical feasibility of the project, outlining the coding environment, software architecture, and database design. Early Figma prototypes have also been discussed, showing that the planned system is user-friendly and achievable in the project's timeframe. Based on this feasibility analysis, the system was implemented using a lightweight web-based architecture, prioritising core gameplay functionality while allowing for future extensibility.

5 Implementation

This section presents the implementation of the system C1PHER based on the design and feasibility considerations outlined in the previous sections. In Section 5.1, the software architecture is detailed. Following on from the architecture is Section 5.2 which outlines the key system components. Section 5.3 discusses the problems encountered and how they were approached and solved. Finally, Section 5.4 summarises the chapter.

5.1Software Architecture

5.1.1Architectural Overview

The C1PHER system was designed using a lightweight client-server architecture. The application is web-based and accessible through a standard web browser. The frontend provides the user interface and manages gameplay interactions, while a Node.js backend handles data management, user accounts, and challenge data retrieval through a RESTful API.

The frontend is responsible for user interaction and game mechanics, whereas the backend manages user authentication, challenge data retrieval, and progress tracking. Progress tracking includes monitoring a user's advancement through the educational journey and calculating the total score awarded for completion of challenges. User data is stored in a JSON file that is dynamically updated as users complete challenges and progress through their journey. In Figure 4, the software architecture of C1PHER can be seen.

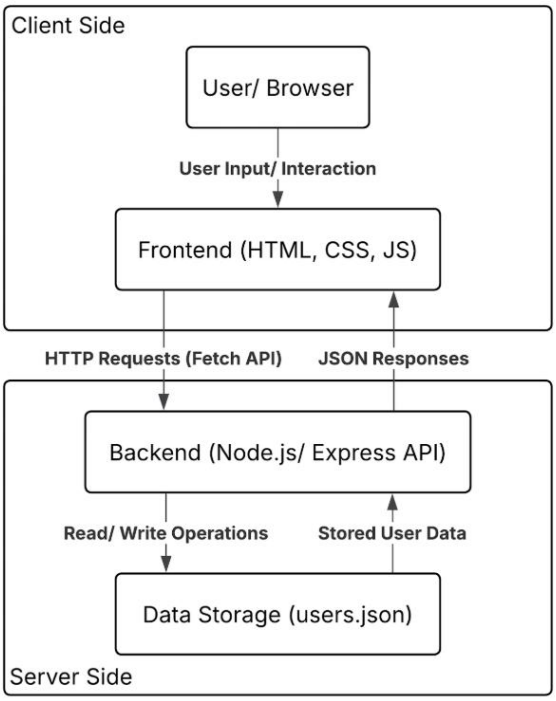

Figure 4: Software Architecture of the C1PHER system

5.1.2Frontend Architecture

The frontend was implemented using HTML, CSS, and JavaScript. This approach aligns with the feasibility considerations outlined in Section 4, as these technologies were already familiar and enabled rapid development without introducing additional complexity from external frameworks.

A central JavaScript file (main.js) manages core client-side functionality, including event handling, and communication with the backend via the Fetch API. The application consists of several interconnected pages, including a login/signup page, journey page, cipher information pages, challenge pages, a profile page, and a settings page (Figure 5). Navigation between pages is handled through standard web routing, with user state maintained using LocalStorage. This provides a lightweight method for keeping the authenticated user across pages without requiring a full session management system.

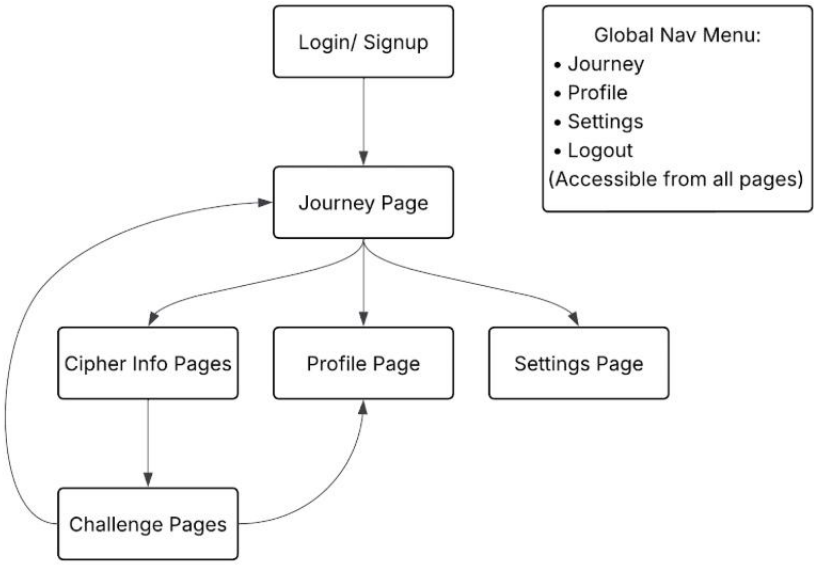

Figure 5: Page flow of C1PHER

5.1.3Backend Architecture

The backend architecture is comparatively simpler than the frontend, and was implemented using Node.js with the Express framework. This was chosen during development due to its lightweight nature and suitability for implementing a RESTful API. This allowed efficient communication between the frontend and backend while keeping the system simple. This provides a RESTful API to support the application's core functionality. Express was selected due to its lightweight nature and suitability for small-scale applications.

The backend is responsible for handling user authentication, retrieving challenge data, and updating user progress. Communication between the frontend and backend follows a request-response model, where HTTP requests are sent from the client and processed by the server, which returns responses in JSON format.

RESTful APIs were selected due to their simplicity, compatibility with HTTP-based web applications, and ability to support scalable client-server communication.

5.1.4API Design and Interaction

The RESTful API enables communication between the frontend web application and the backend server by handling requests related to authentication, challenge retrieval, and progress tracking.

The main API endpoints include:

- /api/login – authenticates user credentials
- /api/register – creates new user accounts
- /api/user/:username – retrieves user profile and progress information
- /api/challenges/:type – retrieves challenge data for a specific cipher challenge
- /api/complete – records completed challenges and updates user scores and progress
- /api/read-page – tracks which information pages have been viewed

These endpoints are handled by backend API routes (Appendix D.7), which process incoming requests and return appropriate responses. This allows the frontend to dynamically retrieve and update data, ensuring that user progress and interactions are consistently reflected across the system.

5.1.5Data Storage Design

User data is stored in a JSON file (users.json), which is accessed and updated using Node.js file system operations. This approach was chosen over the originally proposed MariaDB database to reduce implementation complexity and ensure timely project completion.

While JSON-based storage simplifies development and debugging, it introduces limitations in scalability, security and concurrent access. This represents a trade-off between development efficiency and long-term system robustness. An example of the JSON data structure is provided in Appendix D.8, demonstrating how user credentials, progress, and challenge-related data are represented.

5.1.6Architecture Summary

Overall, the software architecture supports the project's core requirements by enabling dynamic interaction between the user interface and underlying data, while maintaining a clear separation between frontend and backend components.

5.2Core System Components

The C1PHER system is composed of several core components that collectively support its educational and interactive objectives. These components implement the primary functionality of the application, including user management, gameplay mechanics, and learning support features.

5.2.1User Authentication System

The system provides basic user authentication through login and registration functionality. Users can create an account by submitting a username and password, which is then stored in the users.json file. During login, credentials are validated against stored data, and successful authentication allows access to the main application. As

shown in Figure 6, the home page provides a simple interface for user login and registration.

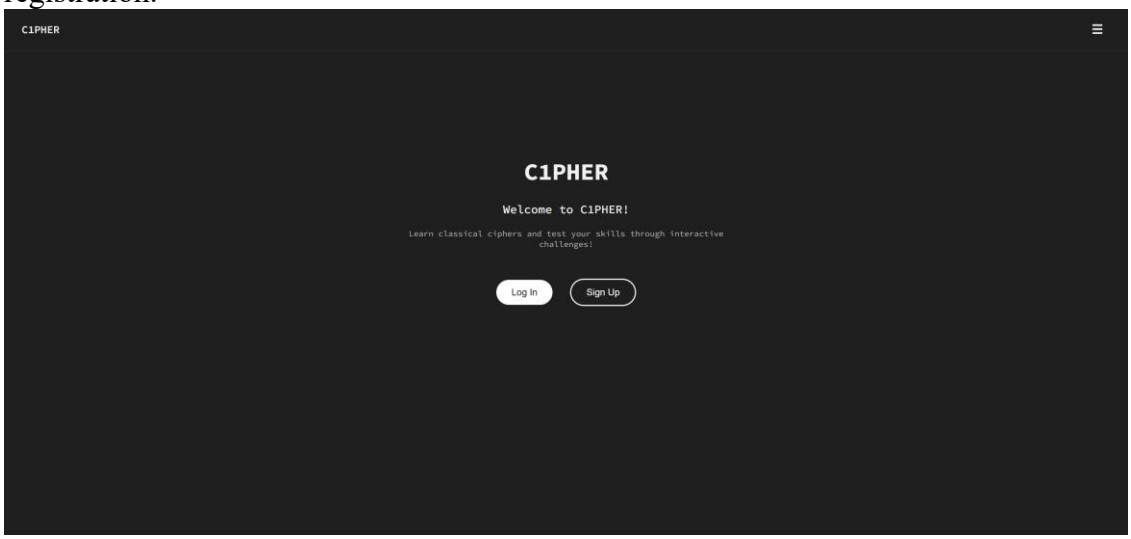

Figure 6: Home page featuring login and signup

The client-side logic for handling login and registration, including communication with the /api/login and /api/register endpoints, is implemented using JavaScript event listeners and Fetch API requests (Appendix D.9).

To maintain user state across pages, the authenticated username is stored in LocalStorage. This enables the system to retrieve user-specific data, such as progress and scores, without requiring repeated authentication requests. Although passwords are currently stored in plain text, this decision was made to prioritise development simplicity. In a full implementation, passwords would be securely hashed and stored in a database to comply with best practices and data protection requirements (NFR2).

5.2.2Game Structure and Navigation

The journey page acts as the central hub of the application and represents the primary gamification element of the system. It presents the available ciphers in a structured chronological format. With more development, features would have been added to indicate visually completed or ongoing ciphers, showing clearer how far through the journey a user has progressed. This design would have supported user motivation and engagement better (RQ1) by providing a clear sense of advancement through the learning material. However, time did not allow for this and so the journey is currently more static in appearance. The current appearance of the journey page can be seen in Figure 7.

Users can navigate from the journey page to individual cipher information pages and corresponding challenges. Completed challenges contribute to the user's overall progress and score, reinforcing continued interaction with the system.

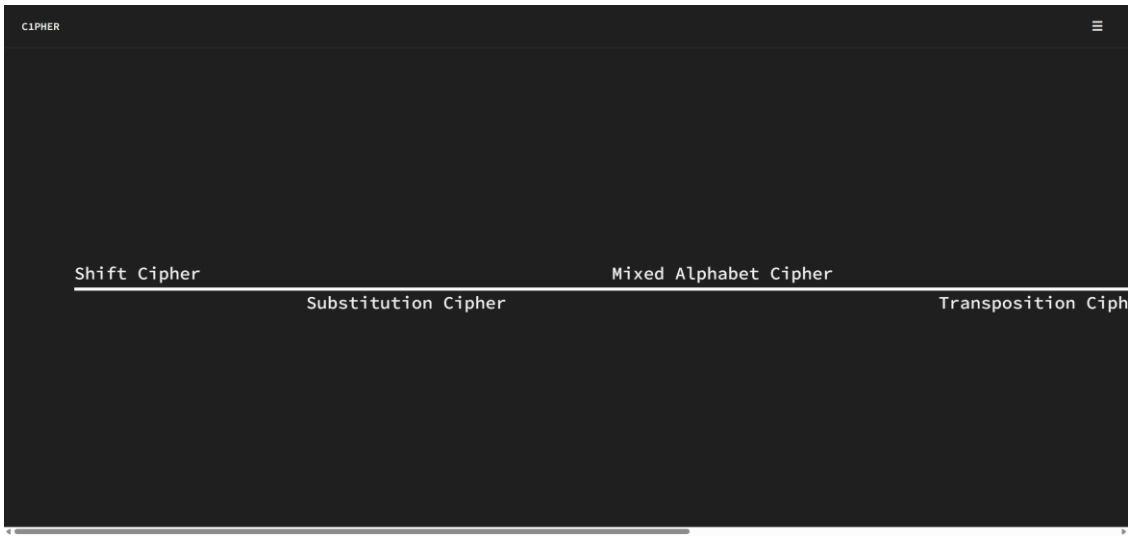

Figure 7: Journey page displaying cipher learning roadmap

5.2.3Educational Content

Each cipher is accompanied by an information page that introduces its underlying concepts, encryption method, and relevant background. These pages serve an educational purpose, ensuring that users understand the theory before attempting challenges. The start of the Shift Cipher information page can be seen in Figure 8 below.

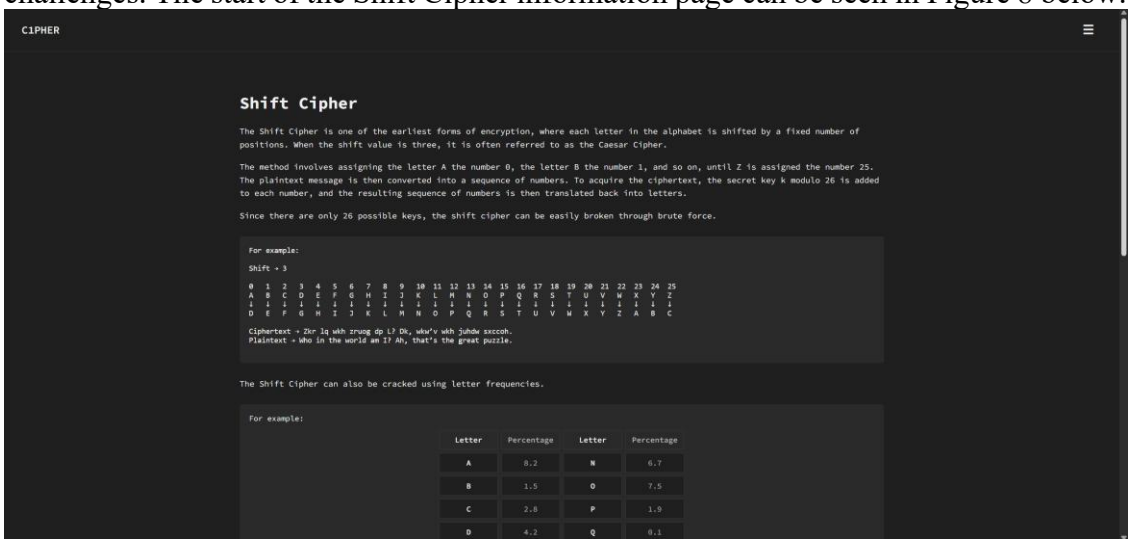

Figure 8: Example information page

These pages directly implement the project's aim of combining learning with gameplay by bridging traditional instructional content with interactive activities. It also supports FR1 by providing clear and accessible explanations of each cipher. As well as having the challenges as interactive activities on the site, some of the ciphers requiring more visual aides for their explanation have interactive visuals on their information pages. One such example is the Rail Fence Cipher, which has an example encryption that can play an animation of encryption and decryption. This can be seen in Figure 9.

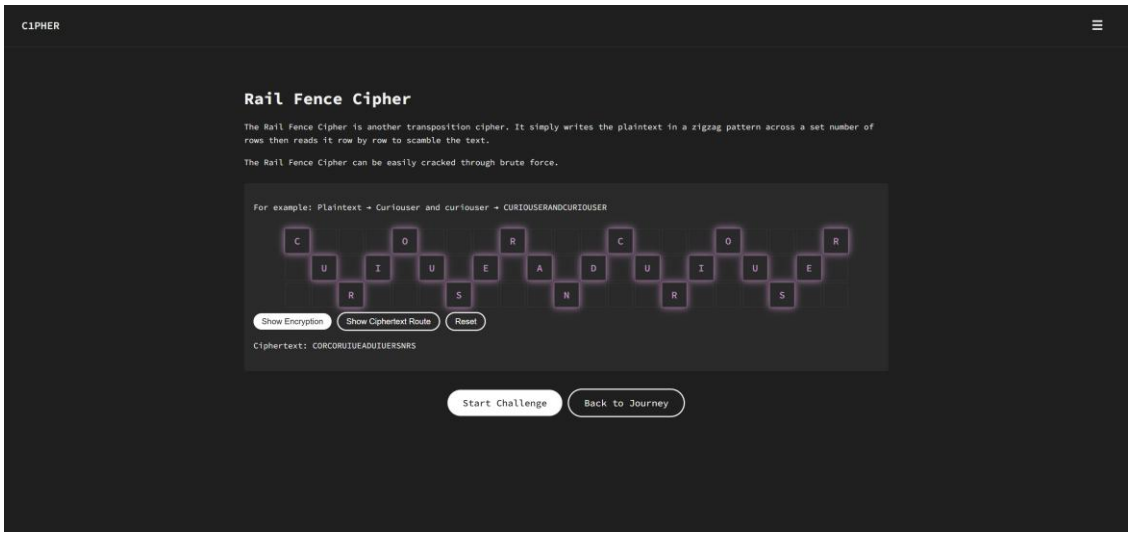

Figure 9: Interactive visual for information page

5.2.4Challenge System

The challenge system forms the core interactive component of the application. Challenges are defined in the backend and retrieved dynamically by the frontend. Each challenge presents a ciphertext that the user must either encrypt or decrypt.

User input is normalised to ensure fairness and usability. Specifically, inputs are standardised by removing spaces, ignoring case differences, and excluding punctuation. This ensures that logically equivalent responses, such as “cipher”, or “CIPHER”, or “C i p h e r” are treated consistently, reducing unnecessary user frustration. This approach improves robustness by preventing valid answers from being incorrectly rejected due to formatting differences. To aid with the text based challenges, a Wordle-style feedback system was introduced. The implementation of this feedback mechanism, including the logic for comparing user input and generating visual indicators, is handled through JavaScript (Appendix D.10). Clicking on the ? button explains how it works and is accessible in all text based challenges as well as having it pop up on the first challenge so that the way the challenge works is explicitly clear to the users. An example of this can be seen in Figure 10. Showing that correct letters, present but in the wrong location and incorrect letters are all indicated. As well as this, all challenges clearly ask for either a certain number of letters or words to be encrypted/ decrypted so that users who are still new to concepts don’t feel frustrated at the concept of having to decode large chunks of text. All of this is in an attempt to encourage the learning experience by adding gamification concepts where possible.

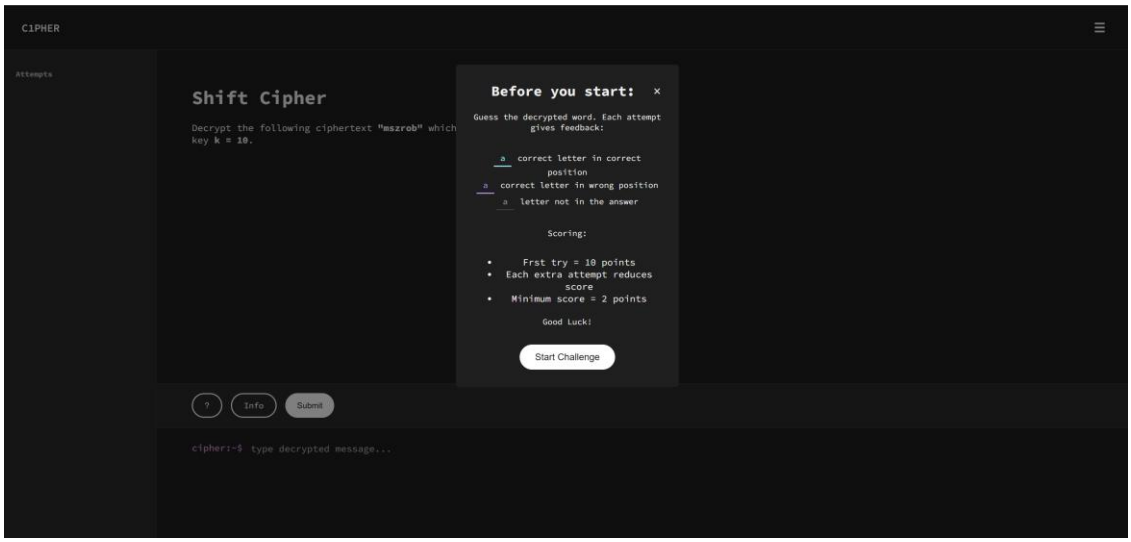

Figure 10: Pop-up explaining wordle-style attempts

Feedback is provided immediately after submission, indicating whether the answer is correct and displaying the users overall score for the challenge. This aligns with FR3 and reinforces learning through instant feedback. On entering a correct answer the popup appears immediately, whereas for incorrect solutions the attempt is logged, showing where users have gone wrong and allows for an immediate next attempt. The correct answer popup can be seen in Figure 11. Should users attempt to submit an answer too short the system alerts them to this with a popup, as well as if they try to submit without any answer. Unlike Wordle there is no cap on attempts and regardless of how many attempts users need to reach the correct answer they will still earn points. This is again another way to encourage users to persevere.

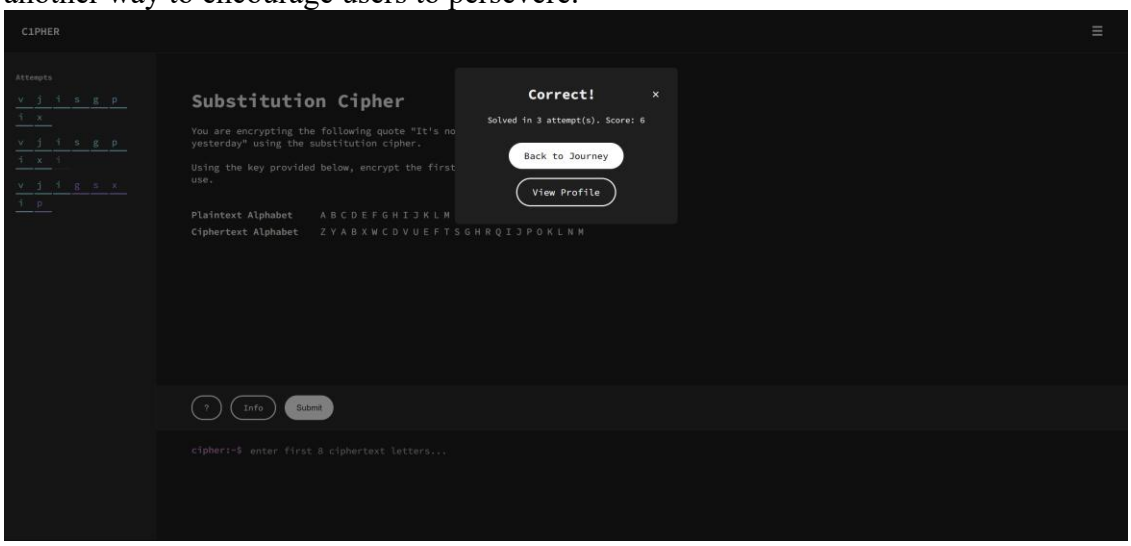

Figure 11: Immediate feedback on challenge completion

5.2.5Hint System

The hint system provides varying levels of support depending on the challenge type. For interactive challenges, such as the Route and Rail Fence ciphers, hints guide users through the solution process step-by-step by revealing the next stage of the solution. This supports structured problem-solving and progressive learning.

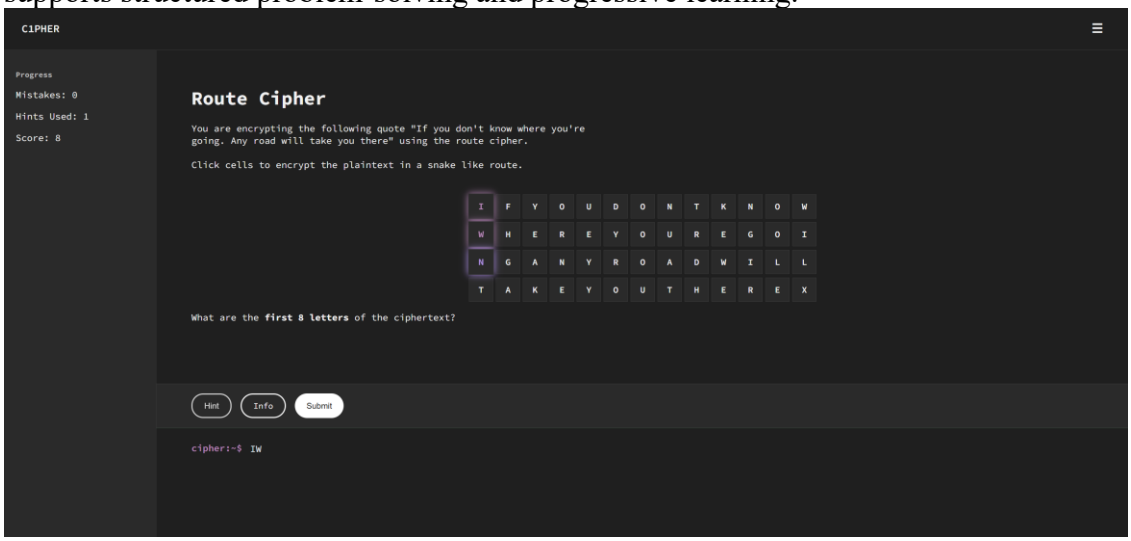

Figure 12: Hint button usage on interactive challenge

In Figure 12 this can clearly be seen as after clicking the hint button, the next letter has been given with a different colour, clearly differentiating from the users current progress through the challenge. This is implemented by identifying the next correct step in the sequence and temporarily highlighting the corresponding grid cell (Appendix D.11). While the same core logic is used for both the Route and Rail Fence challenges, the structures are slightly different.

It can also be seen that if users make mistakes, this also has a clear indicator through colour which also differs from hints and correct selections, as in Figure 13.

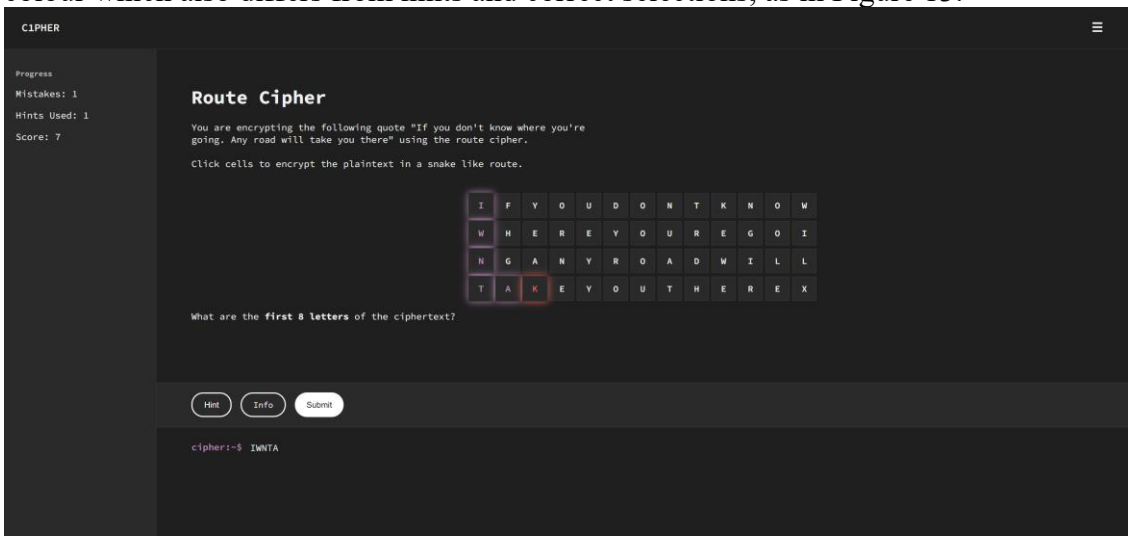

Figure 13: Interactive challenge showing visual feedback for incorrect user input

Incorrect selections are handled by applying a temporary visual indicator to the selected cell (Appendix D.12). Like the hint system, the implementation is consistent for both interactive challenges, with only slight differences, relevant to each cipher.
For text-based challenges, hints provide explanatory guidance rather than direct assistance, encouraging users to engage with the underlying concepts. While this approach supports educational objectives, it highlights an area for future improvement, where more adaptive and personalised hint systems could be introduced. Additionally, users can return to the corresponding cipher information pages at any time, ensuring that sufficient instructional support is available even for users with no prior cryptographic knowledge.

5.2.6Scoring and Progress Tracking

The system includes a scoring mechanism to enhance engagement and provide a sense of achievement. Each completed challenge awards points, with most challenges offering a maximum score of 10 points and more complex challenges, such as the harder Shift Cipher task, awarding 20 points.
User progress is tracked by recording completed challenges and total scores in the backend. This information is displayed in the user's profile and reflected in the journey page. The scoring system supports gamification principles (RQ4) by rewarding progress and encouraging continued participation.

5.2.7User Interface and Accessibility

The user interface was designed to be clear, intuitive, and accessible, supporting objective O5. Consistent layout, clear navigation, and structured content ensure usability across all pages.
A dark mode feature was implemented not only as an accessibility enhancement, but also as a design choice inspired by development environments such as code editors and command-line interfaces. This aligns with the technical nature of cryptography and provides a familiar interface for users. The selected theme persists across sessions, improving usability and personalisation.

5.2.8Error Handling

The system provides user-friendly error messages to guide users when invalid inputs are entered or actions fail. Examples include incorrect login credentials, empty form submissions, and incorrect challenge answers. These messages improve usability by clearly indicating issues and suggesting corrective action, supporting both FR14 and NFR9.

5.2.9Data Management

The system manages user data, challenge completion, and progress using a JSON-based storage approach. This data is stored in the users.json file and includes a user's username, password, points, completed challenges and pages read.
A comprehensive set of screenshots of the system can be found in the Appendix (D.13), these screenshots showcase additional features from the game.

5.3 Problem Solving and Design Decisions

During the implementation of the C1PHER system, several challenges were encountered, requiring design adjustments and trade-offs.

One significant challenge was the initial plan to use a relational database (MariaDB) for data storage. Implementing a full database system introduced additional complexity and time constraints, particularly in relation to schema design, integration, and testing. To address this, the system was simplified by using a JSON file for data storage. While this allowed for rapid development and easier debugging, it introduced limitations in scalability, concurrency, and security. This decision reflects a trade-off between functionality and feasibility within the project's timeframe.

Another challenge involved managing communication between the frontend and backend. Ensuring the API endpoints correctly handled requests and returned appropriate data required extensive debugging, particularly when updating user progress and scores. This was resolved by standardising API responses and carefully structuring request handling.

Maintaining user state across multiple pages also presented difficulties. Without a session management system, tracking the currently logged-in user required an alternative approach. This was addressed using LocalStorage, which provided a simple and effective solution, although it lacks the security of more advanced session handling techniques.

Input validation for challenge answers was another key challenge. Variations in user input formatting, such as differences in case, spacing, or punctuation, could result in correct answers being marked as incorrect. This was resolved by implementing input normalisation, ensuring consistent comparison between user input and expected answers.

Developing the Route and Rail Fence cipher animated examples and the challenge animations also presented several challenges. This process involved significant trial and error, as well as researching existing implementations to better understand how to approach how to go about developing the animations. Specifically, making sure that the order of characters was correct in building the examples. The examples were then used to build the challenges, this introduced issues with making hints and incorrect selections show.

Finally, managing the overall scope of the project required careful prioritisation. While additional features such as a more advanced hint system, expanding challenge sets, and a full database implementation were initially considered, time constraints required a focus on core functionality. This ensured that the primary objectives of the project were achieved while maintaining system reliability.

5.4 Summary of Implementation

This chapter has detailed the implementation of the C1PHER system, including its architecture, core components, and technical decisions. The system successfully integrates educational content with interactive gameplay, supporting the project's aim of making classical cryptography more engaging and accessible. While certain compromises were made, such as the use of JSON for data storage, the implemented

system provides a functional and extensible foundation. The next chapter will evaluate the system's effectiveness in terms of usability, engagement, and learning outcomes.

6 Evaluation

This section evaluates the C1PHER system. Section 6.1 recaps the requirements and details what was and wasn't implemented and why. Section 6.2 discusses the user study conducted and the results achieved. Following this, 6.3 answers the research questions. Section 6.4 presents a critical analysis and discussion of the results acquired. Finally, Section 6.5 summarises the chapter.

6.1Requirements

The system was evaluated against the defined functional and non-functional requirements (section 3.2) to assess implementation completeness and overall system performance.

All of the high-priority functional requirements were successfully implemented fully. The system ended up including six ciphers, all of which had information pages available, as well as at least one challenge for each of these ciphers, with one cipher, the shift cipher having both an easy and a hard challenge. This satisfied FR1, 2, 4 and 6. The easy challenges awarded a maximum of 10 points and the hard challenge awarded a maximum of 20 points, complying with FR7. For the text based challenges, a wordle-style feedback system was implemented, allowing users to see which letters of their guesses were correct, present but in the wrong location, and not present. The interactive challenges had different colour responses depending on whether users had clicked on the right letter or not. On completion of the challenges, pop-ups would appear congratulating users on their success and sharing their score. On all challenge pages, you could also see your score decrease with mistakes as well as hint usage on the interactive challenges. These features allowed for FR3 to be fully implemented. These pop-ups also prompted users if no guesses were made or they were not of the correct length. This complied fully with FR14.

For the medium-priority requirements, two out of the three were partially implemented, with one of them not being implemented at all. For FR5, the optional hints were only implemented on the interactive challenges, allowing users to be given the next letter in the solution. However, for the text based challenges, the user only had the explanation of the wordle-style feedback system and the information pages available and there was no hint system in place allowing users to be given more clues as to how to solve the challenges. FR12 also, was only partially implemented. On the profile page, users could see their progression through the site as a percentage and a total score achieved across the challenges. However, there was no indication of which challenges had been completed or individual scores on challenge history. The final medium-priority requirement was FR11, regarding supporting multiple concurrent users. As the backend was only partially implemented and the user data was stored in a JSON file, there was no support for multiple concurrent users as the JSON file does not support multiple accounts attempting to write to it at the same time.

Of the low-priority functional requirements, only one was partially implemented, the other three were not implemented at all. For FR9, allowing users to register and log in etc., while this was possible, due to the JSON file it was not done in a way that was secure. The data was not encrypted or the passwords hashed. Following on from this, FR10 was not implemented at all, as no external database was set up. FR8, the global leaderboard, would have required a much more extended backend than what was implemented, and so was not able to be implemented given the time constraints. There was also no admin interface setup, to comply with FR13.

Most of the functional requirements not implemented were reliant on a much more extensive backend. The choice was made to focus on core gameplay functionality. That being said, these features remain viable areas for future development and would enhance user engagement and scalability in a full implementation.

Regarding the non-functional requirements, out of the high priority, disappointingly only two out of the six were fully implemented. Due to the small scale of the web game, NFR3 remained reliable and functional without issues through testing. NFR9, the systems error messages, were clear throughout. Any errors with signup, login, or with challenge attempts had clear and user-friendly error messages. NFR6 was partially implemented, for the most part the system seemed to respond within one second or less, however this has been unreliable. This issue only arose once the JSON file for user data was implemented and was not a persistent problem. NFR1, 2 and 4 all related to user data being stored in compliance with GDPR, being stored securely and regular backups taking place as to prevent data loss should a crash occur. These were not implemented, again this comes back to the lack of database. The JSON file did not have any encryption in place and therefore user data was not secure, as well as this, there are no implemented backups in place to protect the data stored.

One out of the three medium-priority requirements were fully implemented, with the third not being implemented. The system was tested using multiple browsers and was functional on all to comply with NFR7. However, the system was unable to be tested on different device types due to requiring commands to start the server, so NFR8 could not be completed. NFR5 could not be implemented, this again links back to the lack of database and user data being stored on a JSON file and so currently cannot scale to handle thousands of users and devices concurrently.

Overall, the system successfully delivers core gameplay functionality, though limitations in backend implementation restrict scalability and security.

6.2System Testing and User Study

The pre-development evaluation stage was not conducted as initially planned. Due to time constraints, the prototype was not developed to a sufficient level to support meaningful user testing. As a result, priority was given to the implementation phase to ensure that core functionality and features were developed to the best of their ability. While this decision limited early user feedback, it allowed for a more complete and functional final system.

For the post-development evaluation, a total of 6 participants took part in the study. The information sheet, consent form and questionnaire given to the participants can be found

in the Appendix (D.14-16). Prior to using the system, five participants reported having some prior knowledge of classical ciphers, although most indicated low confidence in their knowledge. One participant had no prior knowledge, providing a useful baseline for evaluating accessibility.

Participants were divided into three groups (A, B and C), each completing a subset of challenges to ensure coverage of all cipher types without requiring full completion of the game. The challenge split is as follows:

- Group A completed the shift cipher challenges, the substitution cipher challenge and the route cipher challenge.
- Group B completed the shift cipher challenges, the mixed alphabet cipher challenge and the rail fence cipher challenge.
- Group C completed the shift cipher challenges, the transposition cipher challenge and the route cipher challenge.

All participants were asked to create an account and complete the assigned challenges, with the option to consult the cipher information pages if needed.

Performance data indicates that simpler challenges were completed quickly, while more complex tasks required significantly longer completion times. The first shift cipher challenge had an average completion time of 2 minutes and 2 seconds, whereas the more complex letter frequency analysis challenge showed greater variation, with completion times ranging from approximately 2 to 10 minutes and an average of 5 minutes and 22 seconds.

Other challenges, such as substitution, mixed alphabet, and transposition ciphers, were generally completed in under three minutes on average, with some tasks, including the route and rail fence ciphers, completed in under one minute. This suggests that while the system effectively introduces core concepts, certain challenges may lack sufficient difficulty for some users.

Overall performance was high, with an average score of 48 out of 50. This indicates that participants were generally able to complete challenges successfully with minimal errors, reinforcing the system's accessibility but also suggesting limited challenge for more experienced users.

Participants were asked to evaluate the usability of the interface through a series of Likert-scale questions. The statements are as follows:

- 10.1 - The website was easy to navigate
- 10.2 - I was able to understand how the challenges worked without confusion
- 10.3 - The interface felt intuitive to use

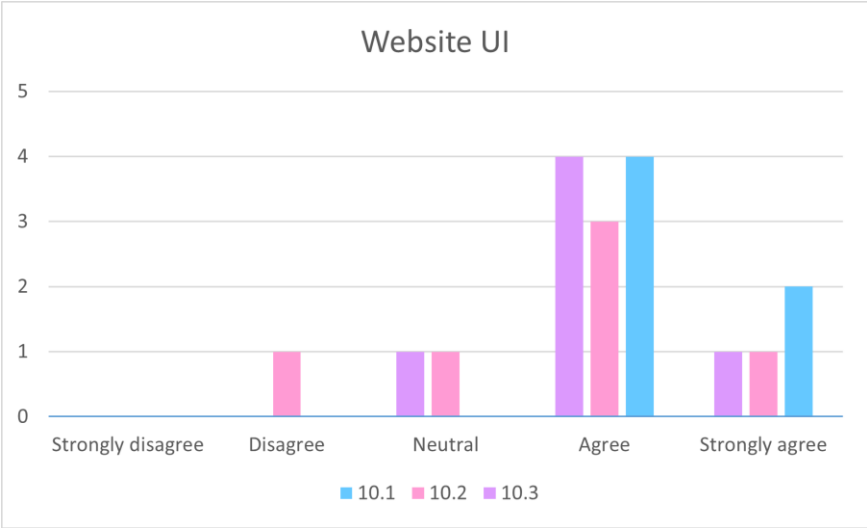

Figure 14: Website UI Feedback Chart

As shown in figure 14, responses were predominantly positive, with most participants agreeing or agreeing strongly that the system was easy to navigate and intuitive to use. A small amount of uncertainty can be seen, with one participant indicating some confusion regarding challenge mechanics. This suggests that while the interface is generally effective, minor improvements to clarity could further enhance usability.
Following on from this, the participants were asked to rate their learning experience, agreeing or disagreeing with the statements below:

- 11.1 - The cipher information pages helped me understand the ciphers
- 11.2 - Completing challenges helped me feel more confident in my understanding of ciphers
- 11.3 - The web game made learning about cryptography more engaging
- 11.4 - I would use this tool to learn classic cryptography over reading books and websites

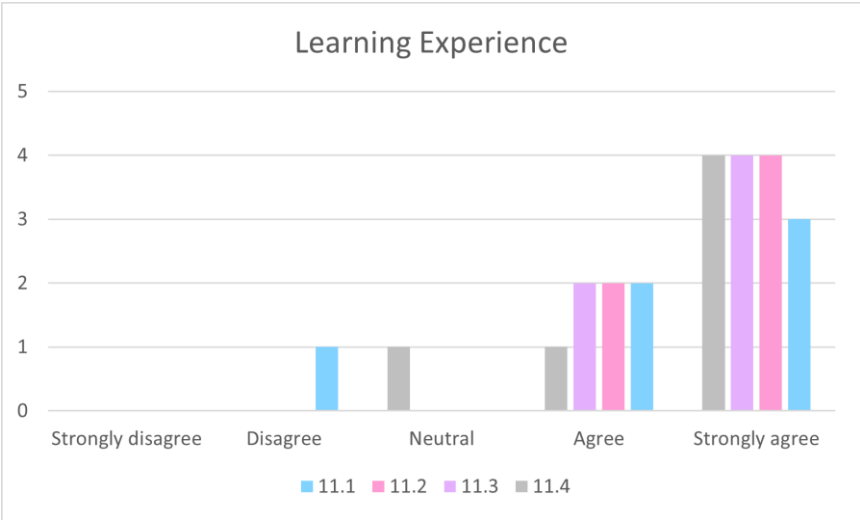

Figure 15: Learning Experience Feedback Chart

As illustrated in figure 15, the majority of participants agreed or strongly agreed that the cipher information pages were helpful and that completing challenges improved their confidence in understanding cryptographic concepts.
Participants also indicated that the game-based approach made learning more engaging compared to traditional methods. Notably, a strong proportion of participants expressed a preference for using the system over textbooks or static resources. This supports the effectiveness of interactive, feedback-driven learning in improving engagement.
One participant disagreed with the usefulness of the information pages, which may reflect prior knowledge rather than a limitation of the system. Overall, the results suggest that the system successfully supports both engagement and perceived learning outcomes.
Lastly the participants were asked to rate the design of the challenges. The statements were as follows:

- 12.1 - The scoring/feedback system was helpful for understanding my progress
- 12.2 - The challenge mechanics (e.g. input, hints, etc) were clear
- 12.3 - The difficulty of the challenges felt appropriate

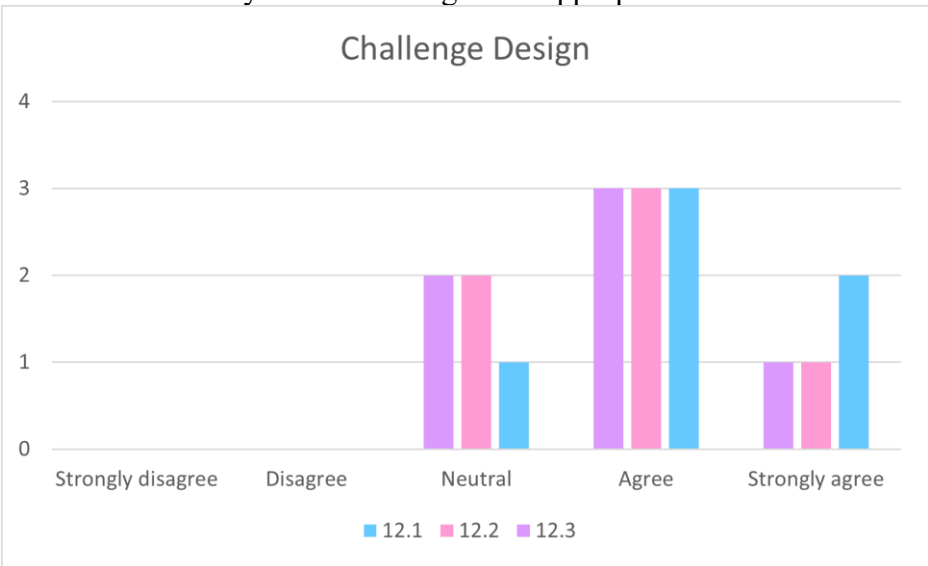

Figure 16: Challenge Design Feedback Chart

Shown above in Figure 16, there was no negative feedback regarding challenge design, and the participants mainly agreed with the statements.
Responses relating to difficulty were generally positive, although slightly more varied. While most participants agreed that the difficulty was appropriate, the performance data suggests that some challenges may be too easy for certain users. This indicates that while the current design is effective for accessibility, expanding the range of difficulty could improve engagement for users with higher prior knowledge.
Qualitative feedback provided additional insights into user experience. One participant noted that it was not immediately clear that additional ciphers were accessible through horizontal scrolling, indicating a minor usability issue. Another comment highlighted readability as an area for improvement.

Positive feedback emphasised the interactive nature of the system, with participants nothing that animations and interactivity made the experience more engaging. Additionally, multiple participants suggested the inclusion of a timer to track completion time, which could enhance both user motivation and performance tracking.

6.3 Research Questions (Section 3.1)

RQ1: How can gamification make the learning of classical cryptography more engaging and accessible to users?

The results indicate that gamification significantly enhances both engagement and accessibility in learning classical cryptography. Participants showed a clear preference for the game-based approach over traditional methods, suggesting that interactive environments are more motivating. Features such as progression, scoring, and immediate feedback provided clear goals and reinforced continued interaction.

The interactive nature of the challenges encouraged active learning, allowing users to apply knowledge and receive instant feedback, which supports understanding and accessibility across different experience levels. However, some participants required additional guidance for more complex challenges, meaning for the game to remain educational, support must be provided in the form of instruction and hints.

RQ2: What combination of cipher complexity and instructional guidance optimally supports users' learning and problem-solving performance?

The findings suggest that an effective balance between challenge complexity and instructional support is essential for learning. Even users with little prior knowledge were able to complete all challenges, demonstrating that the information pages and hints, provided ample support.

However, some challenges were completed very quickly, perhaps suggesting that some challenges were too easy even for inexperienced users. On the contrary, more complex challenges required additional support, highlighting the importance of maintaining guidance alongside increasing difficulty. Expanding the range of challenge complexity would improve both engagement and learning outcomes.

RQ3: Which cipher types and difficulty levels provide the most effective balance between educational value and playability?

A diverse range of cipher types was effective in supporting both educational value and engagement by exposing users to different cryptographic techniques and new challenges. However, difficulty level played a key role in maintaining playability. While simpler challenges supported accessibility, some were perceived as too easy, potentially reducing engagement.

Additionally, most cipher types were represented by a single challenge, limiting progression within each category. Practice is vital for educational value and ensuring users actually retain the information they learn, as well as this, multiple challenges per cipher allows for a range in difficulty. However, care must be taken to avoid monotony, suggesting that an effective balance requires both diversity in cipher types and variation in difficulty.

RQ4: What feedback mechanisms and scoring systems effectively reinforce users' cryptographic understanding and influence user motivation?

Immediate feedback and performance-based scoring were effective in reinforcing understanding and motivating users. Wordle-style feedback in text-based challenges and visual cues in interactive tasks allowed users to iteratively refine their solutions, supporting active problem-solving.

The scoring system encouraged careful engagement by reducing points for mistakes and hint usage, discouraging trial-and-error approaches. However, its simplicity may limit long-term motivation, as it does not account for factors such as completion time. More advanced scoring or adaptive feedback mechanisms could further enhance user engagement.

RQ5: To what extent does the developed game improve users' understanding of classical ciphers compared to traditional static learning materials?

The evaluation suggests that the game improves users' understanding of classical ciphers more effectively than traditional static materials in the short term. Participants reported increased confidence and preferred the interactive approach, indicating that active learning and immediate feedback enhance engagement and comprehension.

However, the findings are limited by the small sample size and evaluation period, meaning long-term learning outcomes were not assessed. While the system demonstrates clear benefits over passive learning methods, further study is required to evaluate its long-term educational impact.

Overall, the findings indicate that the game provides a more engaging and effective learning experience than traditional static materials such as textbooks and lectures.

6.4 Analysis and Discussion of Results

The evaluation of the C1PHER system demonstrates that the integration of gamification, interactive challenges, and immediate feedback can effectively support both user engagement and learning of classical cryptography. However, the results also highlight several limitations and areas for improvement, particularly in relation to system scalability, difficulty balancing, and instructional support.

A key strength of the system is its use of gamification to enhance engagement. Features such as progression through the journey page, scoring mechanisms, and interactive challenges encourage users to remain engaged and complete tasks. This is supported by user feedback, which indicated a preference for the game-based approach over traditional learning methods.

The challenge system was effective in promoting active learning. By requiring users to actively encrypt and decrypt text rather than passively consume information, the system supports a more hands-on approach to learning. Immediate feedback mechanisms, including Wordle-style feedback and visual cues, allow users to iteratively refine their understanding.

Despite these strengths, the evaluation revealed that the current range of challenge difficulty is limited. While simpler challenges were effective for introducing concepts and building user confidence, some participants reported that certain tasks were too easy and completed them very quickly. This suggests that the system may not sufficiently challenge users with prior knowledge or higher skill levels. Despite this, more complex challenges required additional support, often in the form of hints or revisiting the

information pages. These findings indicate that multiple challenges per cipher, increasing difficulty slightly each challenge, would improve learning outcomes as well as engagement.

The information pages provided by the system were effective, allowing users with no prior knowledge to be able to complete even the more complex challenges. Despite all participants in the study being able to complete all challenges asked of them, the lack of prompting and hints for some of the challenges was raised as an issue. Participants with lower confidence or experience occasionally required additional guidance, suggesting that more personalised or context-sensitive hints could improve accessibility and reduce reliance on trial-and-error approaches.

The system successfully delivers its core functionality but is restricted by limitations in backend implementation. The use of a JSON file for data storage simplified development but severely limited scalability, security, and support for concurrent users. This directly impacted several functional and non-functional requirements, particularly those related to data security and system scalability.

The scoring system and feedback mechanisms were effective in reinforcing user understanding and motivation. By linking performance to rewards and penalising excessive use of hints or incorrect attempts, the system encouraged users to engage more thoughtfully with challenges and discouraged rushing and guessing. However, the simplicity of the scoring system may limit its long-term motivational impact. Participants suggested additional features, such as time-based scoring, which could introduce further challenge and competitiveness.

Finally, while the user study provided valuable insights into usability and learning outcomes, it is important to acknowledge its limitations. The small sample size and short duration of the study restrict the relevance of the findings regarding long term usage of an educational game. The evaluation focused primarily on short-term engagement and perceived learning, rather than long-term knowledge retention.

Overall, the results demonstrate that the C1PHER system provides an effective and engaging platform for learning classical cryptography. However, improvements in difficulty progression, instructional support, and backend scalability would further enhance the system's effectiveness and applicability.

6.5Summary of Evaluation

This chapter has discussed the evaluation of the system. All of the high-priority functional requirements were fully implemented and two out of three of the medium-priority requirements were implemented. For the non-functional requirements, two out of six high-priority requirements were fully implemented, the others were not implemented at all due to the lack of database implementation. One out of three of the non-functional requirements were implemented. Despite this the system successfully delivers core gameplay functionality. The user study was overall very positive. The user study allowed the research questions to be discussed and answered. Finally the implementation and study were analysed. These findings are summarised and reflected in the final chapter, which outlines the overall contributions, limitation, and future directions of the project.

7 Conclusion

This section concludes the project. In Section 7.1, the motivations and goals of C1PHER are recontextualised. Following this, section 7.2 details the contributions. Finally, section 7.3 covers the limitations and future work.

7.1 Motivation and Goals

This project was motivated by the challenge of making classical cryptography engaging, as classical ciphers, such as the shift, substitution, and Vigenère ciphers, are fundamental to understanding modern cryptosystems. It aims to design and develop an educational web-based cryptography guessing game that challenges users to identify plaintext messages from ciphertexts encrypted using various classical ciphers. The research focused on exploring classical ciphers and their history, investigating how they can be taught in an engaging and enjoyable environment that makes the learning experience feel like a game rather than a chore.

7.2 Contributions

This project resulted in the successful development of the C1PHER system, a web-based educational game that integrates learning and gameplay to support the understanding of classical cryptography. The system includes a range of cipher types, interactive challenges, and supporting educational content, allowing users to progress through a range of challenges while tracking their performance.

A key contribution of this work is the demonstration that gamification can be effectively applied to technical subjects such as cryptography. Features such as progression through the journey system, performance-based scoring, and immediate feedback mechanisms were shown to enhance user engagement and motivation. In particular, the implementation of Wordle-style feedback for text-based challenges and interactive visual feedback for other challenges provided users with meaningful guidance during problem-solving.

The evaluation of the system further contributes to understanding how different design choices influence learning outcomes. The results indicate that a balance between challenge complexity and instructional support is essential for maintaining both accessibility and engagement. Simpler challenges were effective in introducing concepts and building confidence, while more complex challenges encouraged deeper understanding, particularly when supported by hints and explanatory content.

Additionally, the findings suggest that interactive, feedback-driven learning environments can provide a more engaging alternative to traditional static materials, supporting active learning and improving user confidence. Overall, this project illustrates how gamified systems can support the learning of technical concepts such as classical cryptography.

7.3 Limitations and Future Work

Despite the strengths of the C1PHER system, several limitations were identified during development and evaluation. One of the primary limitations is the user of a JSON file

for data storage, which restricts scalability, security, and support for concurrent users. This prevented the implementation of several functional and non-functional requirements, including secure data handling, database integration, and support for multiple users interacting with the system simultaneously.

Another limitation is the range of challenge difficulty. While the current implementation provides a structured progression from simpler to more complex ciphers, the challenges tend to be introductory level for each, with some participants reporting that certain challenges were too easy, while others required additional guidance. This indicates that a wider range of difficulty levels, including multiple challenges per cipher type, would improve both engagement and learning outcomes.

The hint system also presents an area for improvement. Although hints were effective in supporting users during interactive challenges, they were not consistently implemented across all challenge types. Future work could focus on developing more adaptive and context-sensitive hint systems to better support users with varying levels of prior knowledge.

From an evaluation perspective, the study was limited by a small sample size and short duration, which restricts the generalisability of the findings. Future research could involve larger-scale user studies and longer-term evaluation to assess knowledge retention and learning effectiveness over time.

Further development of the system would most definitely include the integration of a secure database, implementation of a global leaderboard, expansion of challenge content, and the introduction of additional features such as time-based scoring or adaptive difficulty. These enhancements would improve both the scalability and educational value of the system.

Overall, while the C1PHER system demonstrates the potential of gamified learning in the context of classical cryptography, it also highlights opportunities for further refinement and extension.

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Appendices

A.Project Management

The project was predominantly managed through weekly meetings with my supervisor where my work from that week was discussed. The deadlines for ethics and deliverables one and two were used to help me gauge personal deadlines for each section of the project. Every other meeting with my supervisor was a group meeting with other students which provided the opportunity to discuss design aspects for my project and receive multiple opinions. After meetings I would make notes of any feedback provided and use it to help guide my workload for the next week. GitLab was used to share research and some incremental work with my supervisor.

A.1. Gantt Chart

1. Semester 1 Gantt Chart

Cryptography guessing game

Semester 1

Megan Brennan

TASK	PROGRESS	START	END
1 Background & Motivation			
1.1 Define aims & objectives	100%	22-Sep-25	25-Sep-25
2 Background research			
2.1 Cryptography overview	100%	22-Sep-25	27-Sep-25
2.2 Classical Ciphers	100%	27-Sep-25	02-Oct-25
2.3 Modern Cryptography	100%	01-Oct-25	07-Oct-25
2.4 Gamification and Educational Tool	100%	05-Oct-25	13-Oct-25
2.5 Research write-up	100%	10-Oct-25	22-Oct-25
2.6 Summary	100%	20-Oct-25	26-Oct-25
3 Work Plan			
3.1 Research Goals	100%	27-Oct-25	28-Oct-25
3.2 Requirements	100%	28-Oct-25	29-Oct-25
3.3 Tasks	100%	29-Oct-25	30-Oct-25
3.4 Evaluation strategy	100%	30-Oct-25	31-Oct-25
4 Feasibility			
4.1 Coding environment	100%	03-Nov-25	06-Nov-25
4.2 Design	100%	06-Nov-25	10-Nov-25
4.3 Prototypes	100%	10-Nov-25	16-Nov-25
5 Conclusion			
5.1 Motivation & Goals	100%	10/11/25	12/11/25
5.2 Proposed work	100%	13/11/25	16/11/25
6 D2 & Ethics proposal			
6.1 Ethics proposal	100%	13/10/25	26/10/25
6.2 Semester 2 plan	100%	17/11/25	19/11/25
6.3 Video creation	100%	19/11/25	20/11/25

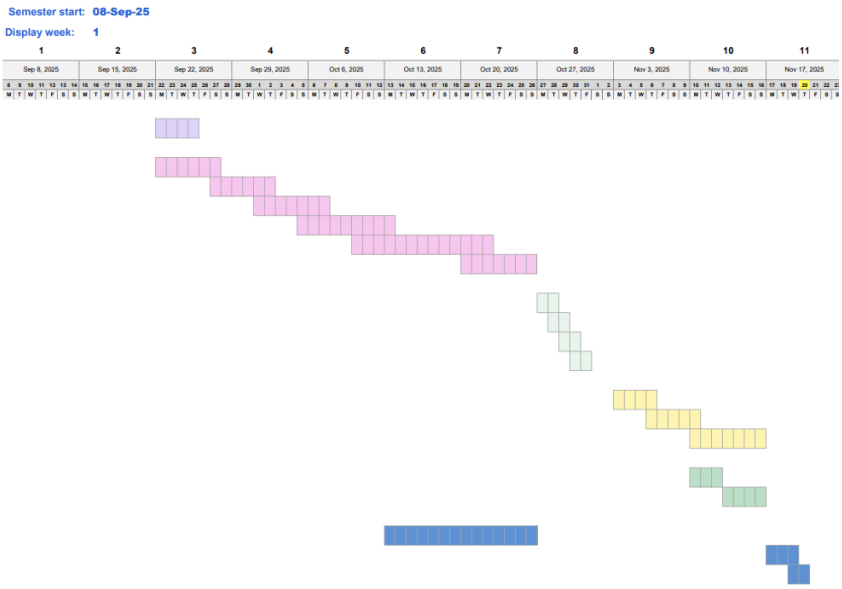
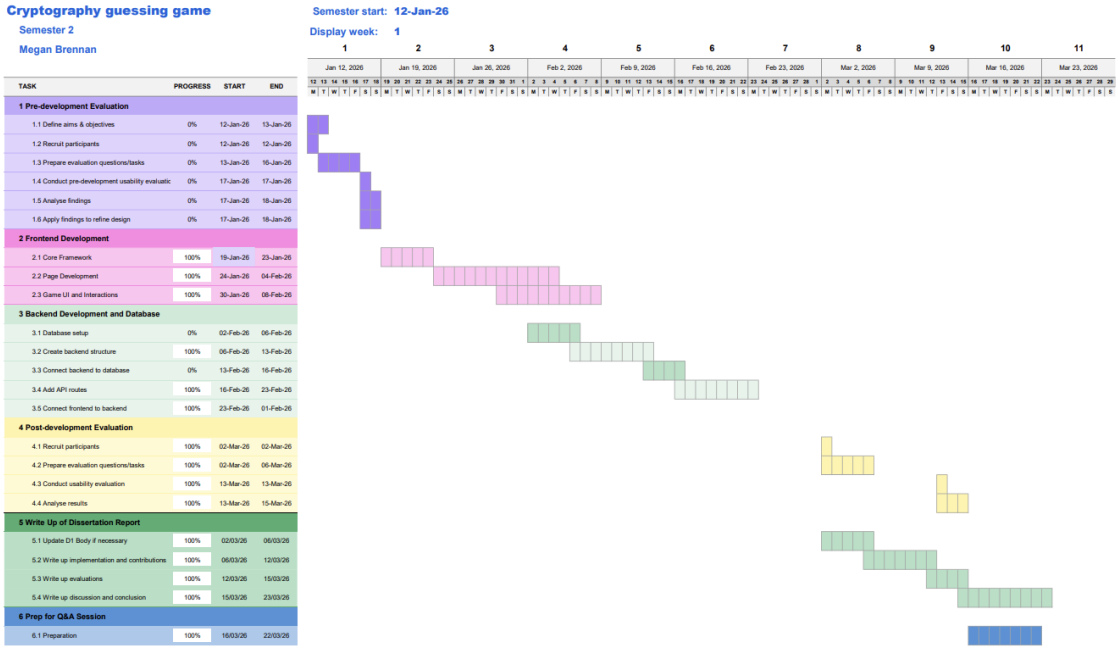


Figure 17: Gantt Chart Semester 1

2. Semester 2 Gantt Chart



Cryptography guessing game
Semester 2
Megan Brennan

Semester start: 12-Jan-26
Display week: 1

TASK	PROGRESS	START	END
1 Pre-development Evaluation			
1.1 Define aims & objectives	0%	12-Jan-26	13-Jan-26
1.2 Recruit participants	0%	12-Jan-26	12-Jan-26
1.3 Prepare evaluation questions/tasks	0%	13-Jan-26	16-Jan-26
1.4 Conduct pre-development usability evaluatic	0%	17-Jan-26	17-Jan-26
1.5 Analyse findings	0%	17-Jan-26	18-Jan-26
1.6 Apply findings to refine design	0%	17-Jan-26	18-Jan-26
2 Frontend Development			
2.1 Core Framework	100%	19-Jan-26	23-Jan-26
2.2 Page Development	100%	24-Jan-26	04-Feb-26
2.3 Game UI and Interactions	100%	30-Jan-26	08-Feb-26
3 Backend Development and Database			
3.1 Database setup	0%	02-Feb-26	06-Feb-26
3.2 Create backend structure	100%	06-Feb-26	13-Feb-26
3.3 Connect backend to database	0%	13-Feb-26	16-Feb-26
3.4 Add API routes	100%	16-Feb-26	23-Feb-26
3.5 Connect frontend to backend	100%	23-Feb-26	01-Feb-26
4 Post-development Evaluation			
4.1 Recruit participants	100%	02-Mar-26	02-Mar-26
4.2 Prepare evaluation questions/tasks	100%	02-Mar-26	06-Mar-26
4.3 Conduct usability evaluation	100%	13-Mar-26	13-Mar-26
4.4 Analyse results	100%	13-Mar-26	15-Mar-26
5 Write Up of Dissertation Report			
5.1 Update D1 Body if necessary	100%	02/03/26	06/03/26
5.2 Write up implementation and contributions	100%	06/03/26	12/03/26
5.3 Write up evaluations	100%	12/03/26	15/03/26
5.4 Write up discussion and conclusion	100%	15/03/26	23/03/26
6 Prep for Q&A Session			
6.1 Preparation	100%	16/03/26	22/03/26

Figure 18: Gantt Chart Semester 2

A.2. Analysis of Risks

This risk analysis overviews potential project hazards that could compromise the development or successful completion of the project. Each risk analysis contains type specification, probability level and type level. The risk description defines the risk, and the mitigation strategy suggests possible solutions.

R1: Underestimation of overall development time
Description: The time required to implement all planned features (front end, backend, challenges, user evaluation) may be greater than initially estimated.
Risk type: Project
Probability: Medium
Impact: High
Indicators: Progress on early tasks is slower than planned, repeated need to adjust schedule.
Strategy (Mitigation): Review the project timeline weekly, reduce non-essential features early, and prioritise core functionality before enhancements.

R2: Difficulty implementing backend features
Description: Implementing backend features such as user accounts, scoring, or data storage may take longer than estimated, potentially exceeding the planned schedule.
Risk type: Technical
Probability: Medium

Impact: High
Indicators: Inability to complete backend components during early development milestones, repeated delays or unresolved errors.
Strategy (Mitigation): Prioritise essential backend functionality and reduce non-critical features, create a simplified or mock backend if necessary.

R3: Copyright issues
Description: Images, fonts, or cipher explanations sourced online may be copyrighted, creating legal or academic integrity issues.
Risk type: Legal
Probability: Low
Impact: Medium
Indicators: Use of assets not explicitly labelled as free, unclear licensing.
Strategy (Mitigation): Use open-source or licence-free assets only and provide full citations for any external material.

R4: Accessibility shortcomings
Description: The interface may not meet accessibility expectations, making the game difficult for users with visual or cognitive impairments.
Risk type: Social/Ethical
Probability: Medium
Impact: Medium
Indicators: Small text, insufficient contrast, unclear navigation, lack of alternative input methods.
Strategy (Mitigation): Follow WCAG guidance, where possible offer dark/light modes, and test design with accessibility checkers.

R5: Improper handling of user data (GDPR concerns)
Description: If a login system is implemented, collected data such as usernames or progress could introduce GDPR responsibilities.
Risk type: Legal/Ethical
Probability: Low
Impact: High
Indicators: Storing any personal data without a clear retention policy or justification.
Strategy (Mitigation): Minimise or avoid storing personal data, if necessary, store only pseudonymised identifiers and provide clear privacy information.

A.3. PLES

Professional Issues: The project should follow the BCS Code of Conduct, ensuring that professional standards are upheld throughout. Complete documentation of all design decisions, algorithms, and implementation steps have been or will be clearly documented to maintain high professional standards. All external sources used should be accurately cited and referenced to ensure academic integrity.

Legal Issues: Any storage of personal data will comply with UK GDPR alongside the Data Protection Act 2018. The project also avoids using any copyrighted images or assets without permission, ensuring that all external materials are properly licensed or referenced.
Ethical Issues: The project ensures fairness, transparency, and respect for users. Any user accounts will store only necessary information and will protect user privacy. Feedback, scoring, and game mechanics are designed to avoid misleading or harming users. Accessibility features, including readable text and clear navigation, help ensure that the system is usable for a wide range of learners.
Social Issues: Complete documentation of all design decisions, algorithms, and implementation steps have been or will be clearly documented so that it is understandable to future developers. Socially, the system contributes to increasing public engagement with cryptography by presenting it in an approachable, game-based format. This helps lower barriers for individuals who may find the subject difficult or intimidating, promoting broader digital literacy. Accessibility features, such as readable text and clear navigation ensure that users with varying abilities can participate.

B. Generative AI

Generative AI was used to help structure my background research, breaking it down into subheadings. GenAI was also used to help find some of the articles used, for example the papers on AES and Twofish. It was also used to help bridge the gap between wanting to produce a game and break it down into feasible ideas. For chapters 5-7, AI was used to help breakdown the chapters into sections. Finally, GenAI was used to check spelling, grammar and tone. No AI generated content was included in the final submission.

C. Changes since Research Report

The introduction was built on to include the details of the sections implementation and evaluation as well as the new conclusion that has been added. As well as this, sections 2, 3 and 4 have had some changes made based on the feedback provided last semester. The modern cryptography section was cut down to allow for space to explore the use of gamification in education. Other than this there have only been small changes like slight wording changes or adding short sentences to link sections slightly better.

D. Additional Material

D.1. English letter frequencies [3]

Table 3: English Letter Frequencies

Letter	Percentage	Letter	Percentage
A	8.2	N	6.7
B	1.5	O	7.5
C	2.8	P	1.9

D	4.2	Q	0.1
E	12.7	R	6.0
F	2.2	S	6.3
G	2.0	T	9.0
H	6.1	U	2.8
I	7.0	V	1.0
J	0.1	W	2.4
K	0.8	X	0.1
L	4.0	Y	2.0
M	2.4	Z	0.1

D.2. English bigram frequencies [3]

Table 4: English Bigram Frequencies

Bigram	Percentage	Bigram	Percentage
TH	3.15	HE	2.51
AN	1.72	IN	1.69
ER	1.54	RE	1.48
ES	1.45	ON	1.45
EA	1.31	TI	1.28
AT	1.24	ST	1.21
EN	1.20	ND	1.18

D.3. Most common trigrams in decreasing order [3]

Table 5: Most Common Trigrams

THE	ING	AND	HER	ERE	ENT	THA	NTH	WAS	ETH	FOR

D.4. Vigenère Square [22]

Table 6: Vigenère Square

	A	**B**	**C**	**D**	**E**	**F**	**G**	**H**	**I**	**J**	**K**	**L**	**M**	**N**	**O**	**P**	**Q**	**R**	**S**	**T**	**U**	**V**	**W**	**X**	**Y**	**Z**
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y

D.5. Database Architecture

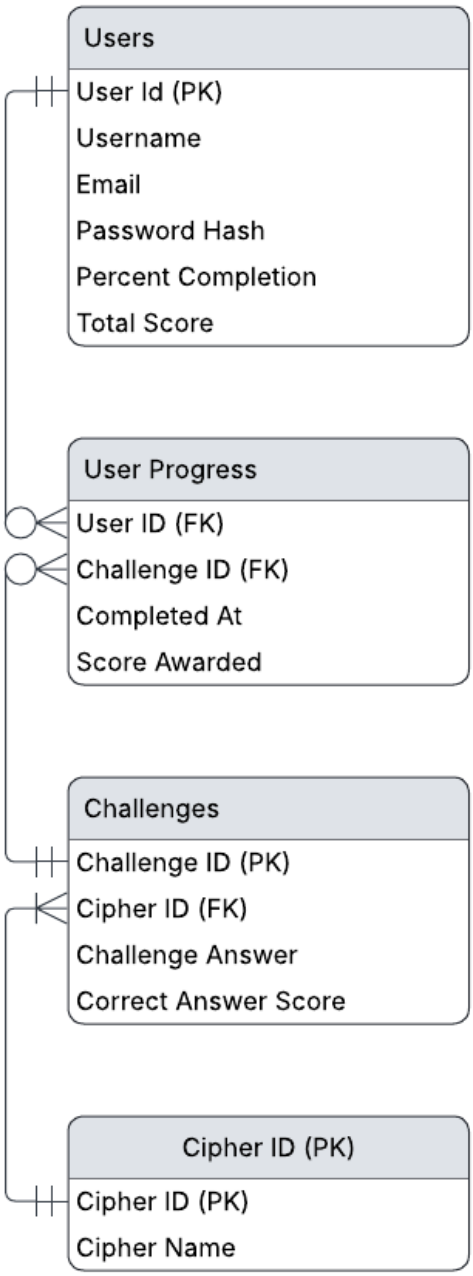


Figure 19: Database Architecture

D.6. Prototype Screenshots

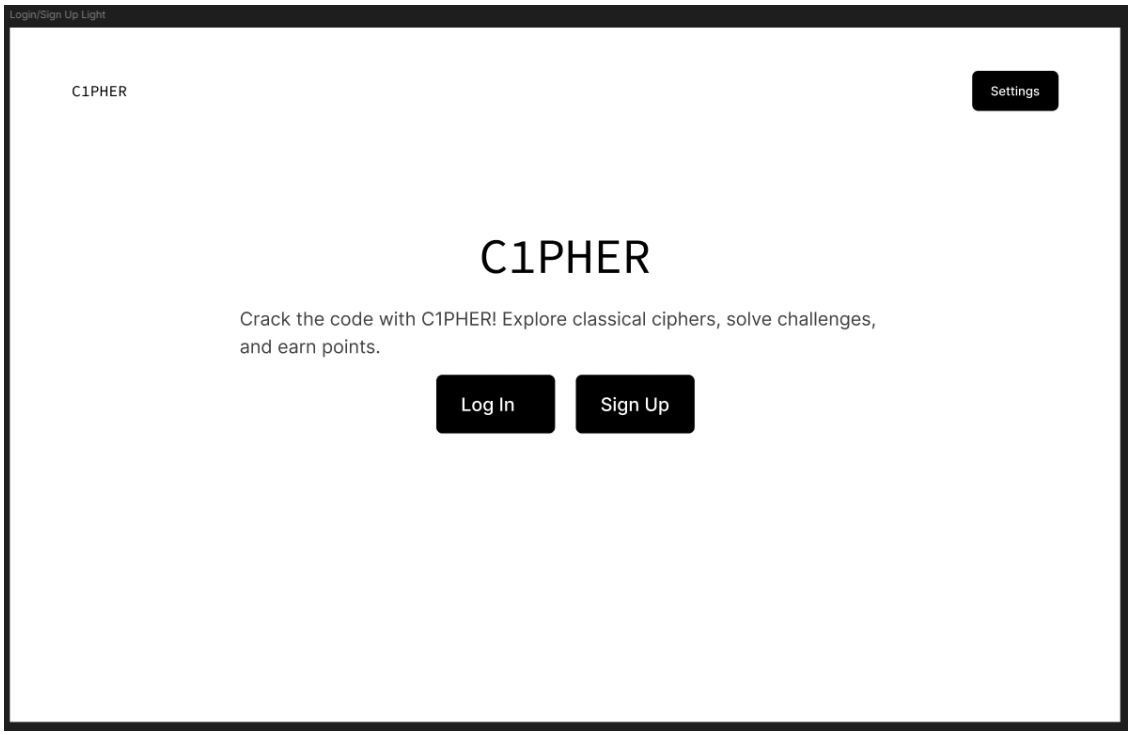


Figure 20: Login/ Sign Up Page (Light)

Figure 21: Login/ Sign Up Page (Dark)

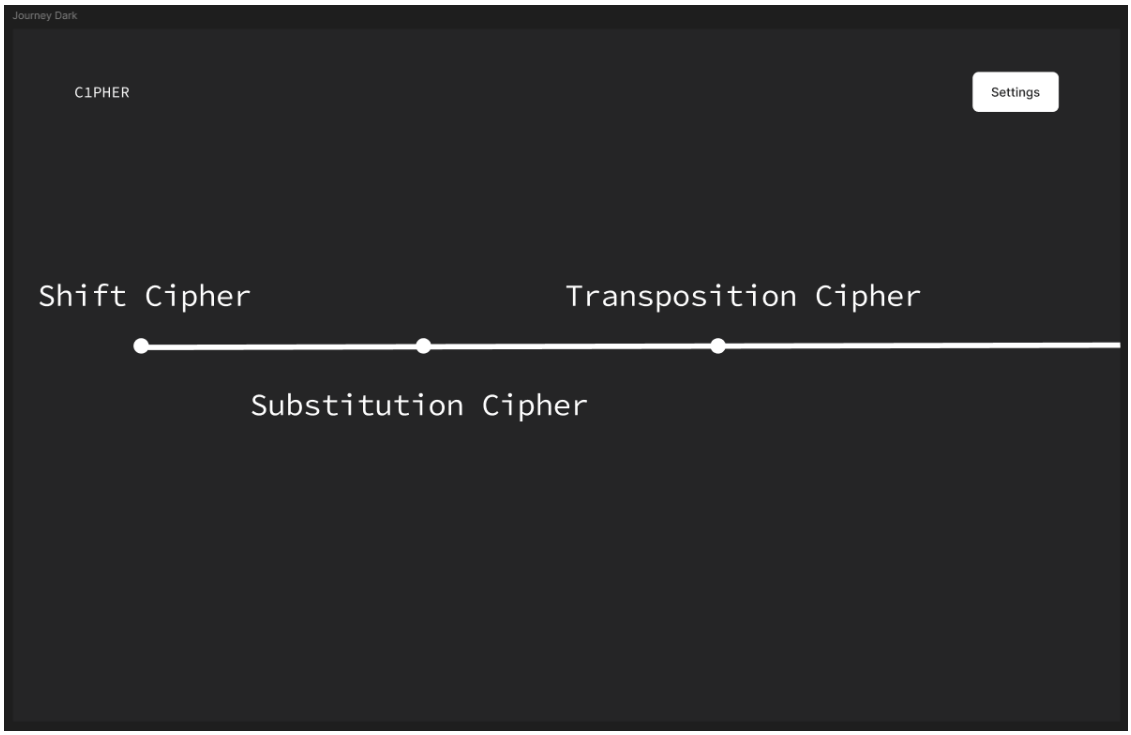


Figure 22: Journey Page (Dark)

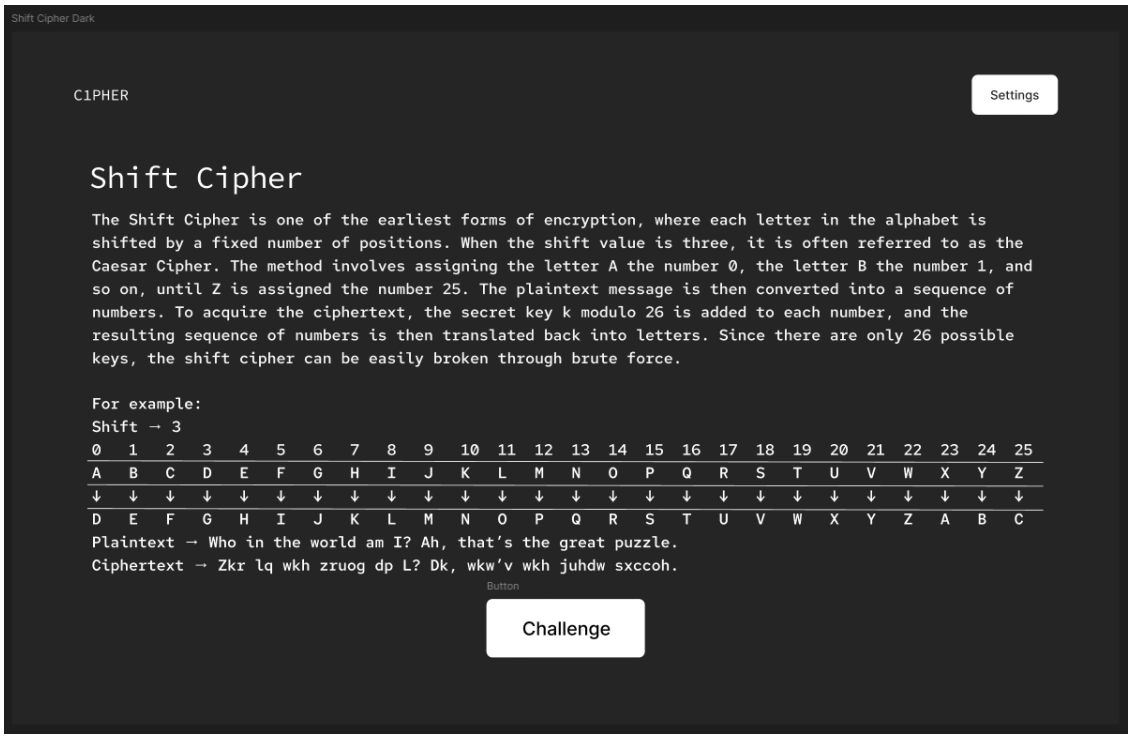


Figure 23: Explanation Page (Dark)

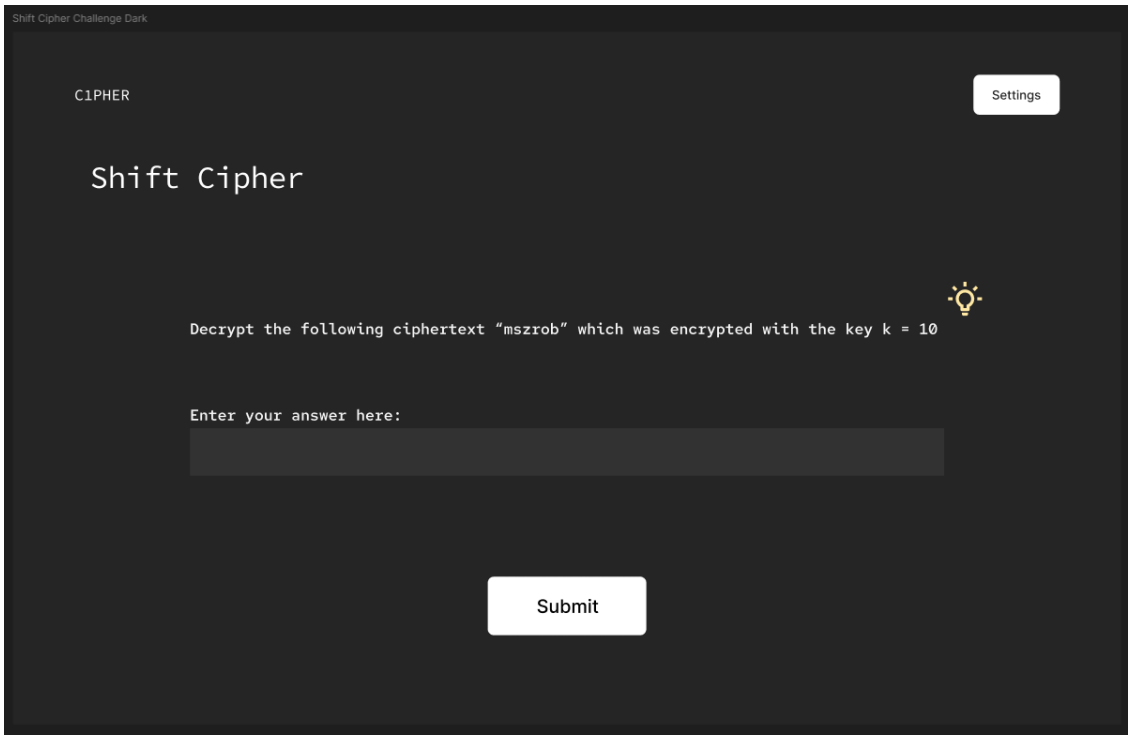


Figure 24: Challenge Page (Dark)

D.7. API Routes

```
/*-----------------------------------------------------------------------------------------
------*/
// API ROUTES
/*-----------------------------------------------------------------------------------------
------*/

/*-----------------------------------------------------------------------------------------
------*/
// AUTHENTICATION ROUTES
/*-----------------------------------------------------------------------------------------
------*/

// POST /api/login
// Authenticates a user using username and password
// Returns success: true if credentials match an existing account

app.post("/api/login", (req, res) => {
    const { username, password } = req.body;
```

```
    const data = loadUsers();
    // Find user with matching username and password
    const user = data.users.find(
        u => u.username === username && u.password === password
    );
    if (!user) {
        return res.json({ success: false });
    }
    res.json({ success: true });
});

// POST /api/register
// Creates a new user account and stores it in users.json
// Usernames are noralised to lowercase to avoid duplicates

app.post("/api/register", (req, res) => {
    const { username, password } = req.body;
    const data = loadUsers();
    // Normalise username (remove spaces and convert to lower case)
    const usernameClean = username.trim().toLowerCase();
    const existingUser = data.users.find( u => u.username === usernameClean);
    // Check if user already exists
    if (existingUser) {
        return res.json({ success: false, message: "User already exists" });
    }
    // Create new user object
    const newUser = {
        username: usernameClean,
        password,
        points: 0,
        completedChallenges: [],
        pagesRead: []
    };
    // Save new user
    data.users.push(newUser);
    saveUsers(data);
    res.json({ success: true });
```

```
});

/*-----------------------------------------------------------------------------------
------*/
// USER MANAGEMENT ROUTES
/*-----------------------------------------------------------------------------------
------*/

// DELTE /api/user/:username
// Deletes a user account from users.json
// Allows users to delete their accounts

app.delete("/api/user/:username", (req, res) => {
    const { username } = req.params;
    const data = loadUsers();
    data.users = data.users.filter(u => u.username !== username);
    saveUsers(data);
    res.json({ success: true });
});

// GET /api/user/:username
// Retrieve profile into for a specific user
// Returns username, score and overall progress

app.get("/api/user/:username", (req, res) => {
    const { username } = req.params;
    const data = loadUsers();
    const user = data.users.find(u => u.username === username);
    if (!user) {
        return res.status(404).json({ error: "User not found" });
    }
    // Calculate overall learning progress
    const totalChallenges = Object.keys(challenges).length;
    const totalPages = 6;
    const progress =
        (user.completedChallenges.length + user.pagesRead.length) /
        (totalChallenges + totalPages);
```

```
    res.json({
        username: user.username,
        points: user.points,
        progress: progress
    });
});

/*-----------------------------------------------------------------------------------
------*/
// CHALLENGE ROUTES
/*-----------------------------------------------------------------------------------
------*/

// GET /api/challenges/:type
// Retrieves challenge data for a specific cipher

app.get("/api/challenges/:type", (req, res) => {
    const { type } = req.params;
    // Checks if requested challenge exists
    if (!challenges[type]) {
        return res.status(404).json({ error: "Challenge not found" });
    }
    res.json(challenges[type]);
});

// POST /api/coomplete
// Records completion of a challenge and updates the users score
// Prevents duplicate completions for the same challenge

app.post("/api/complete", (req, res) => {
    const { username, challengeId, score } = req.body;
    const data = loadUsers();
    const user = data.users.find(u => u.username === username);
    if (!user) {
        return res.status(404).json({ error: "User not found" });
    }
    // prevent duplicate completions
```

```
    if (!user.completedChallenges.includes(challengeId)) {
        user.completedChallenges.push(challengeId);
        user.points += score;
        saveUsers(data);
    }
    res.json({ success: true });
});

// POST /api/read-page
// Tracks when a user reads an info page
// Used for calculating learning progress

app.post("/api/read-page", (req, res) => {
    const { username, page } = req.body;
    const data = loadUsers();
    const user = data.users.find(u => u.username === username);

    if (!user) {
        return res.status(404).json({ error: "User not found" });
    }

    // Only add the page if it hasn't been read before
    if (!user.pagesRead.includes(page)) {
        user.pagesRead.push(page);
    }

    saveUsers(data);

    res.json({ success: true });
});
```

D.8. JSON Data Structure

```
{
  "users": [
    {
      "username": "admin",
      "password": "password",
```

```
      "points": 10,
      "completedChallenges": [
        1
      ],
      "pagesRead": [
        "shift-info"
      ],
      "challengeAttempts": {}
    },
```

D.9. JavaScript Login and Signup Form Implementation

```
/*-----------------------------------------------------------------------------------------
------*/
// LOGIN FORM
/*-----------------------------------------------------------------------------------------
------*/

    const loginForm = document.getElementById("login-form");

    if (loginForm) {
        loginForm.addEventListener("submit", (e) => {
            // Prevents page refresh on submit
            e.preventDefault();

            // Retrieve entered username and password
            const username = document.getElementById("login-
username").value.trim().toLowerCase();
            const password = document.getElementById("login-password").value;

            // Send login request to backend API to authenticate user credentials
            fetch("http://localhost:3000/api/login", {
                method: "POST",
                headers: {
                    "Content-Type": "application/json"
                },
                body: JSON.stringify({
                    username: username,
                    password: password
```

```
                })
            })
            .then(res => res.json())
            .then(data => {

                if (data.success) {
                    // Store logged in user
                    localStorage.setItem("user", username);
                    // Redirect
                    window.location.href = "journey.html";

                } else {
                    alert("Invalid username or password");
                }
            })
            .catch(err => console.error("Login failed:", err));
        });
    }

/*-----------------------------------------------------------------------------------
------*/
// SIGNUP FORM
/*-----------------------------------------------------------------------------------
------*/

    const signupForm = document.getElementById("signup-form");

    if (signupForm) {
        signupForm.addEventListener("submit", (e) => {
            e.preventDefault();

            const username = document.getElementById("signup-
username").value.trim().toLowerCase();
            const password = document.getElementById("signup-password").value;

            fetch("http://localhost:3000/api/register", {
                method: "POST",
```

```
            headers: {
                "Content-Type": "application/json"
            },
            body: JSON.stringify({
                username: username,
                password: password
            })
        })
        .then(res => res.json())
        .then(data => {
            if (data.success) {
                // Store logged in user
                localStorage.setItem("user", username);
                // Redirect to journey page
                window.location.href = "journey.html";
            } else {
                alert(data.message);
            }
        })
        .catch(err => console.error("Signup failed:", err));
    });
}
```

D.10. Wordle-Style Feedback System

```
/*-----------------------------------------------------------------------------------------
------*/
// WORDLE STYLE ATTEMPTS
// Initialises the logic for wordle-style gameplay. The user submits guesses which are
compared
// against the correct answer. Feedback is provided using coloured indicators showing:
// - correct letter in correct position
// - correct letter in wrong position
// - incorrect letter
// The number of attempts impacts the final score awarded to the user.
/*-----------------------------------------------------------------------------------------
------*/

    function initWordleChallenge({
```

```
    challengeId,
    answer,
    maxScore = 10,
    answerInput,
    feedbackArea,
    submitBtn,
    resultModal,
    resultTitle,
    resultMessage
}) {

    // Track number of attempts made by user
    let attempts = 0;

    // Main function: runs when user submits a guess
    function handleGuess() {

        // Stops if anything missing
        if (!answerInput || !feedbackArea || !answer) return;

        // Normalise user input
        const guess = answerInput.value
        .replace(/[^a-zA-Z]/g, "") // removes spaces
        .trim()
        .toLowerCase();

        if (!guess) return;

        // Increase attempt counter
        attempts++;

        // New row for wordle style feedback
        const row = document.createElement("div");
        row.classList.add("feedback-row");

        const guessLetters = guess.split("");
        const answerLetters = answer.split("");
```

```
const spans = [];

// First pass: check for correct letters and position
guessLetters.forEach((letter, index) => {
    const span = document.createElement("span");
    span.classList.add("letter");
    span.textContent = letter;

    if (letter === answerLetters[index]) {
        span.classList.add("correct");
        answerLetters[index] = null;
        guessLetters[index] = null;
    }
    spans.push(span);
});

// Second pass: check for correct letters and wrong position
guessLetters.forEach((letter, index) => {
    if (letter === null) return;

    const span = spans[index];
    const presentIndex = answerLetters.indexOf(letter);

    if (presentIndex !== -1) {
        span.classList.add("present");
        answerLetters[presentIndex] = null;
    } else {
        span.classList.add("absent");
    }
});

// Append letters to feedback row
spans.forEach(span => row.appendChild(span));

// Adds newest attempt to sidebar
feedbackArea.prepend(row);
```

```
            // If correct then popup with score
            if (guess === answer) {
                let score = maxScore - (attempts - 1) * 2;
                if (score < 2) score = 2;

                recordChallengeCompletion(challengeId, score);

                resultTitle.textContent = "Correct!";
                  resultMessage.textContent = `Solved in ${attempts} attempt(s). Score:
${score}`;

                resultModal.style.display = "block";
            }

            // Clears input box for next guess
            answerInput.value = "";
        }

        // Submit button click
        if (submitBtn) {
            submitBtn.addEventListener("click", handleGuess);
        }

        // Allow enter key submission
        if (answerInput) {
            answerInput.addEventListener("keydown", (e) => {
                if (e.key === "Enter") {
                    handleGuess();
                }
            });
        }
    }
```

D.11. Hint System For Interactive Challenges

```
// Hint button
        document.getElementById("hint-btn")
        ?.addEventListener("click", () => {
```

```
            const cells = routeGrid1.querySelectorAll("div");
            // Determine the next correct cell in route
            const hintIndex = correctRoute[playerRoute.length];

            if (hintIndex === undefined) return;

            // Increase hint counter
            hintsUsed++;
            // Update score
            updateScore();

            const hintCell = cells[hintIndex];

            // Highlight hint cell
            hintCell.classList.add("route-hint-active");

            // Remove hint after delay
            setTimeout(() => {
                if (!hintCell.classList.contains("route-selected")) {
                    hintCell.classList.remove("route-hint-active")
                }
            }, 1200);
        });
```

D.12. Wrong Input Feedback For Interactive Challenges

```
// Wrong cell selected
                    else {

                        // Increase mistake counter
                        mistakes++;
                        // Update score
                        updateScore();

                        // Add visual feedback
                        cell.classList.add("route-wrong");
                        cell.offsetHeight;
```

```
        // Remove wrong indicator after delay
        setTimeout(() => {
            cell.classList.remove("route-wrong");
        }, 800);
    }
```

D.13. C1PHER Screenshots

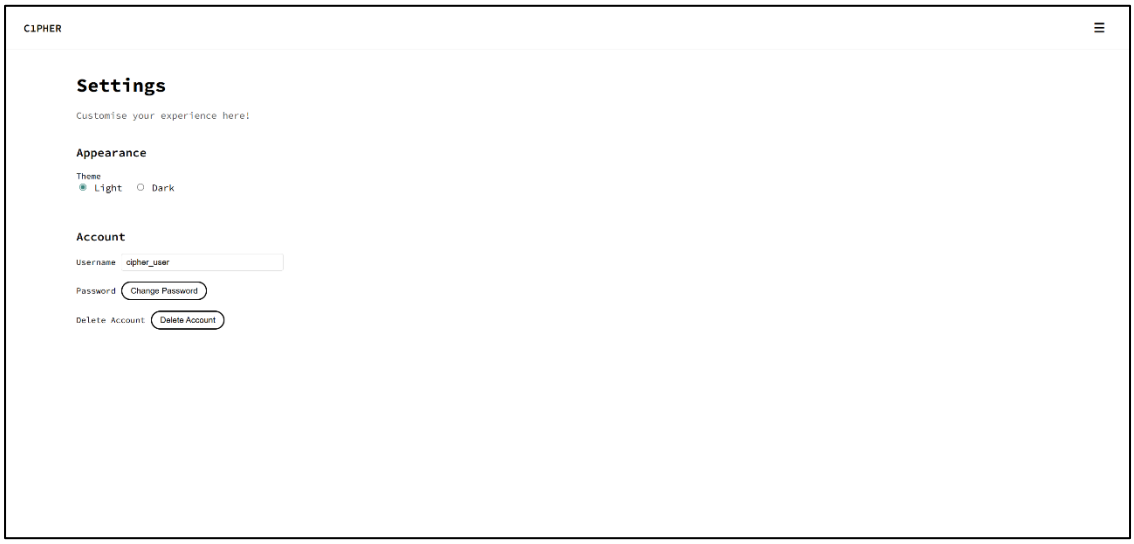


Figure 25: Settings page showing light and dark mode radio buttons

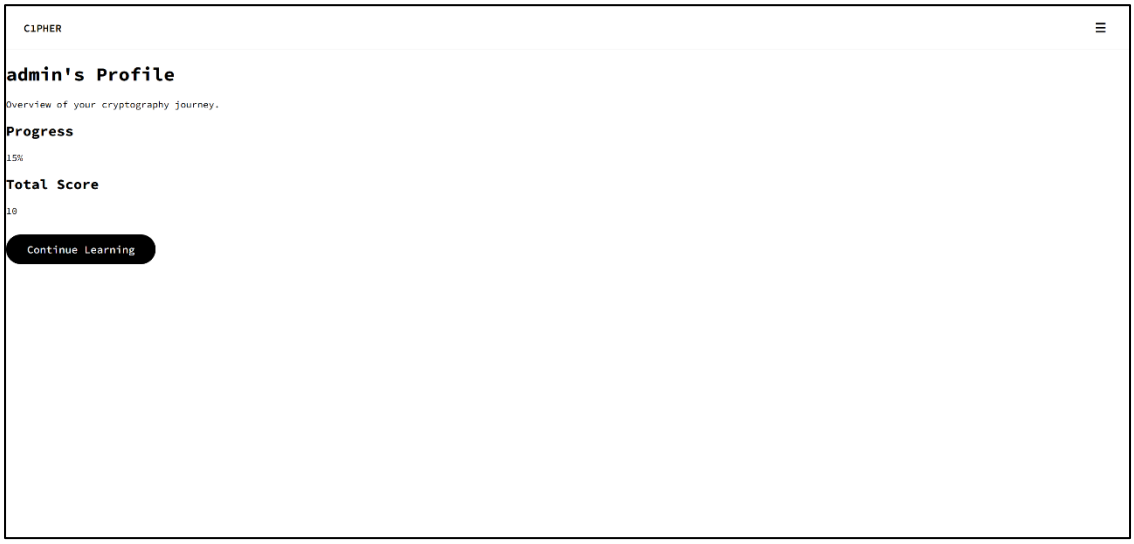


Figure 26: Profile page showing progress and total score

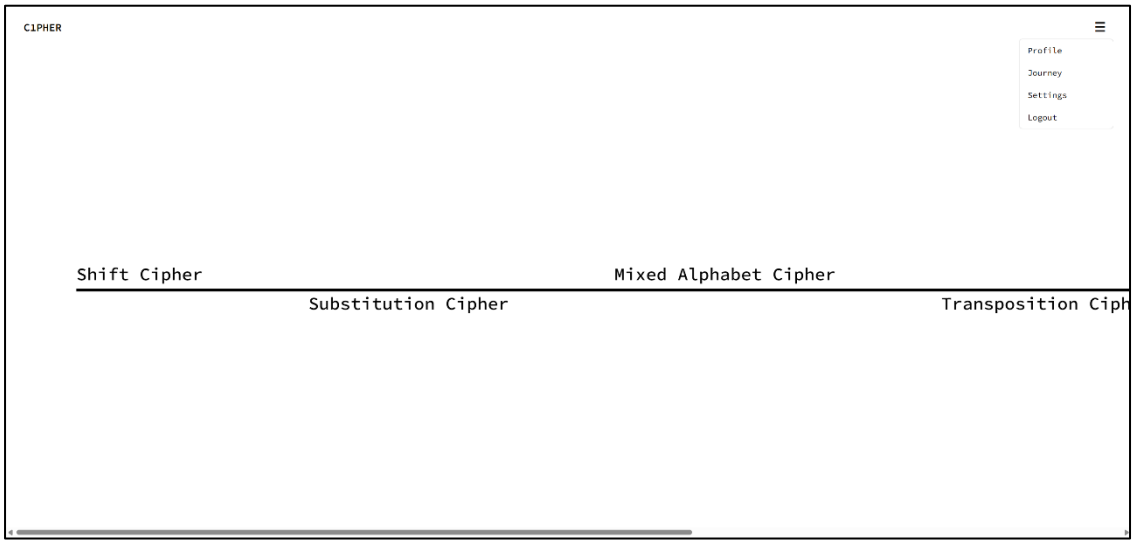
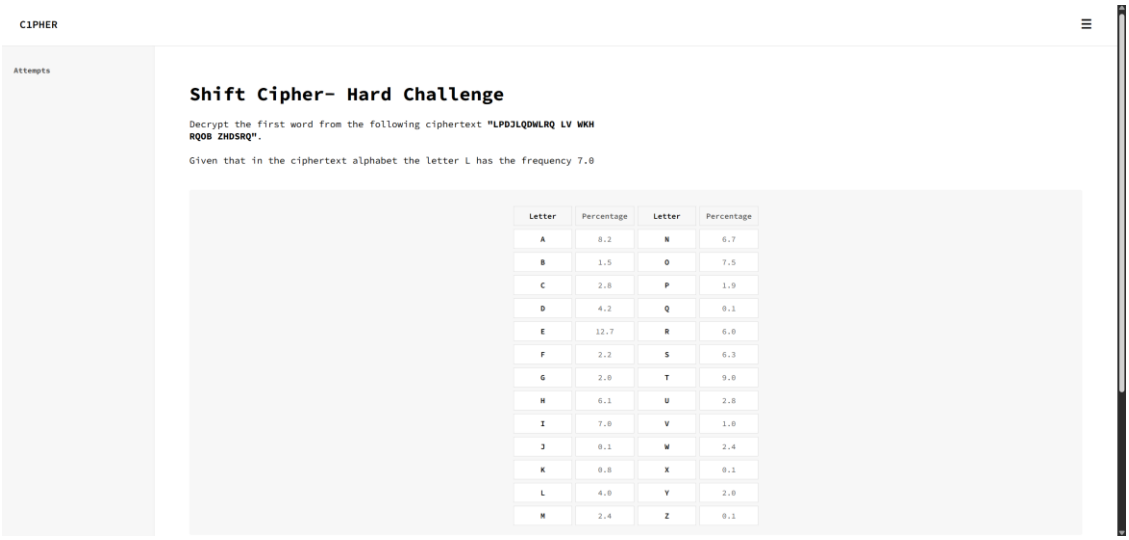


Figure 27: Journey page including drop down menu



Letter	Percentage	Letter	Percentage
A	8.2	N	6.7
B	1.5	O	7.5
C	2.8	P	1.9
D	4.2	Q	0.1
E	12.7	R	6.0
F	2.2	S	6.3
G	2.0	T	9.0
H	6.1	U	2.8
I	7.0	V	1.0
J	0.1	W	2.4
K	0.8	X	0.1
L	4.0	Y	2.0
M	2.4	Z	0.1

Figure 28: Shift cipher hard challenge

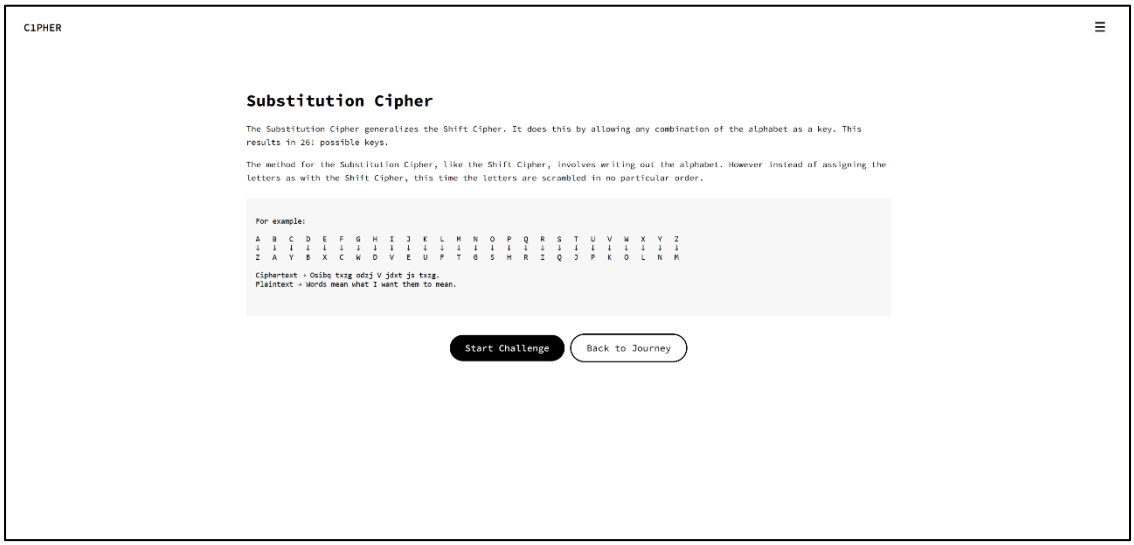


Figure 29: Substitution cipher information page

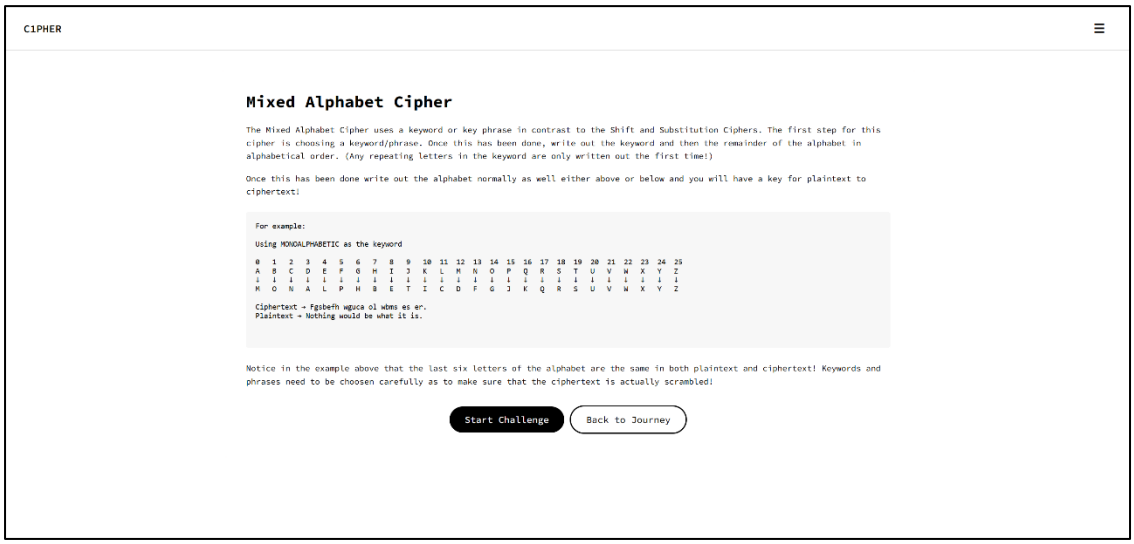


Figure 30: Mixed alphabet cipher information page

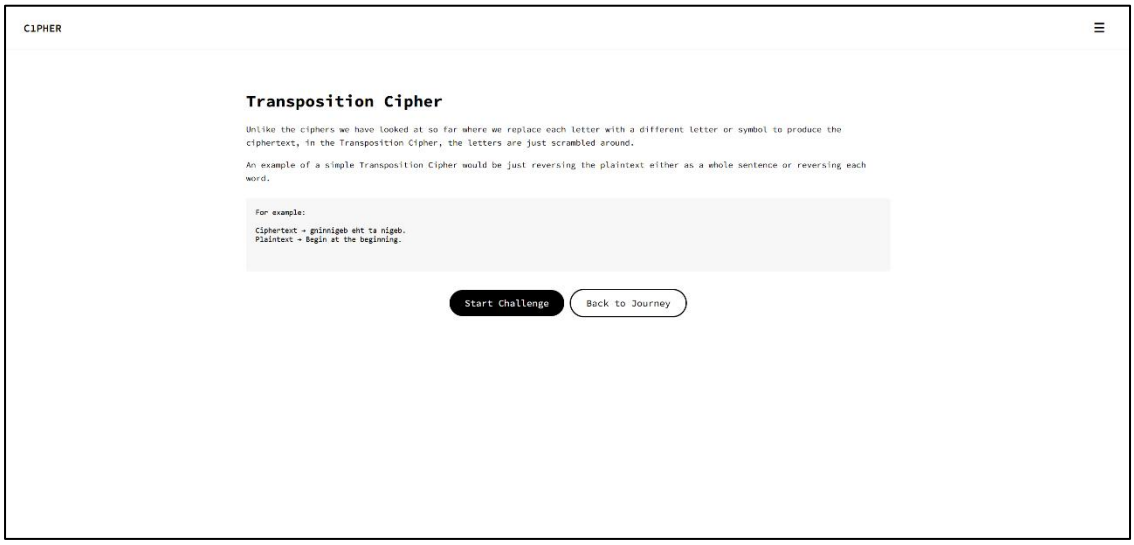


Figure 31: Transposition cipher information page

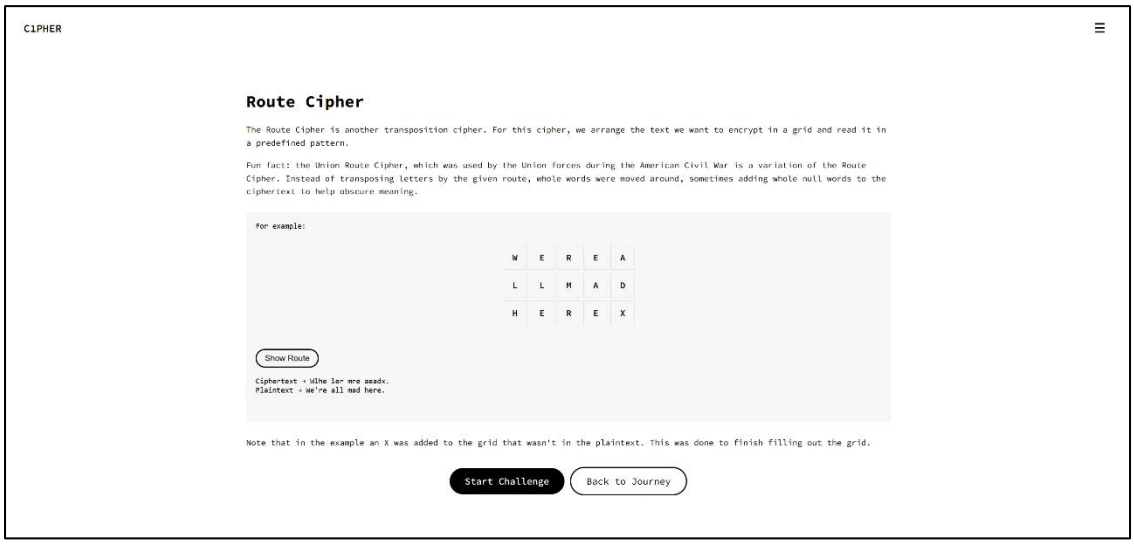


Figure 32: Route cipher information page

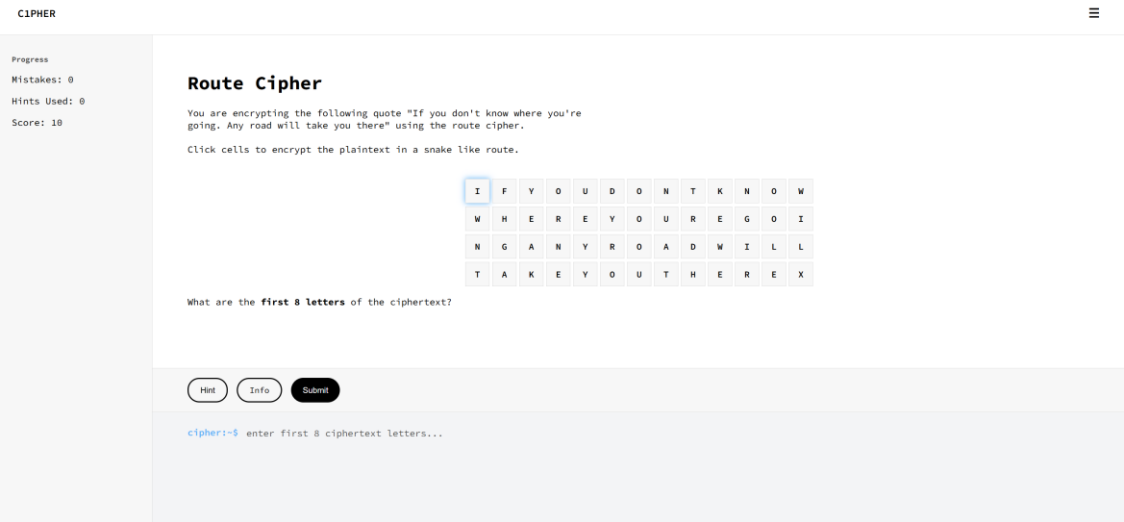


Figure 33: Route cipher challenge

D.14. Information Sheet

HERIOT WATT UNIVERSITY

INFORMATION SHEET

Project title: Cryptography Guessing Game

Objectives

For my Hons dissertation project, Cryptography Guessing Game, I am investigating the effectiveness of gamification on learning classic cryptography.

The aim of this study is to evaluate a web-based cryptography guessing game that helps users learn about classical ciphers. The game will allow users to attempt to encrypt and decrypt messages using classical ciphers. The study seeks to assess the usability, accessibility, and educational effectiveness of the system.

Data collection

The data collected is listed below:

- Confirmation of the participant being aged 18 years or older.
- Confirmation of the participant being enrolled at Heriot-Watt as a Computer Science student in fourth year.
- Feedback from the questionnaires.

The data being collected is pseudo-anonymised. Participants can access or delete their data by sending an email requesting this to either myself (mb2091@hw.ac.uk) or my project supervisor (m.just@hw.ac.uk).

Experiment Protocol

The experiment will take place in person at a time and place organised between myself and the participant. The participant is expected to read through and play some parts of the web game. They can also browse through the rest of the game should they choose to. The tasks asked of the participants will be to carry out four of the challenges included in the game and fill out a questionnaire. The challenges should probably take around 15-20 minutes; however this will depend on the participants knowledge of classical ciphers, and the questionnaire will take around 5-10 minutes to complete.

All data collected will be stored on the Heriot-Watt One Drive ensuring that it is kept secure.

If a participant chooses to participate in this experiment, please complete the consent form and send it to mb2091@hw.ac.uk.

Withdraw

Participants have the right to withdraw at any time before, during or after the experiment without any harm to their student status or grades.

If you wish to take part, please complete the consent form sent to you.

After you have completed the consent form, should you wish to take part, please email mb2091@hw.ac.uk to organise a time and place to carry out the experiment.

Please note that your signed consent form will be securely stored in the HWU One-Drive folder, which is password protected and can be accessed only by the MACS Director of Ethics and HWU Data Protection Officer.

For additional details or to request the deletion of your data, you can reach out to mb2091@hw.ac.uk or you can contact the project supervisor via m.just@hw.ac.uk.

NOTE: Please find below further information on Heriot-Watt Data Protection

1) Data Protection Officer contact details: dataprotection@hw.ac.uk

2) Here is the ink to the Heriot-Watt University Privacy Notice for Research Participants: https://www.hw.ac.uk/uk/services/docs/information-governance/PrivacyNoticeResearch-V4Finalversion.pdf

3) Data controller is Heriot-Watt University

D.15. Consent Form

HERIOT WATT UNIVERSITY

CONSENT FORM

Project title: Cryptography Guessing Game

Heriot-Watt University attaches high priority to the ethical conduct of research. We therefore ask you to consider the following points before signing this form. Your signature confirms that you are willing to participate in this study, however, signing this form does not commit you to anything you do not wish to do, and you are free to withdraw your participation at any time.

Please tick the initial boxes

- **I confirm** that the purpose of this study was explained to me in sufficient detail and I have had the opportunity to consider the information, to ask questions and have these answered satisfactorily. ☐
- **I confirm** that I am over 18 years old. ☐
- **I understand** that my participation is voluntary and that I am free to withdraw at any time, without giving any reason, and without consequences. ☐
- **I understand** that any data I give will be used in the dissemination of findings and will remain anonymous and that my signed consent form will be securely stored in the HWU One-Drive folder, which is password protected and can be accessed only by the MACS Director of Ethics and HWU Data Protection Officer. ☐

- **I feel** I am well enough, physically and mentally, to complete the tasks as described in the Information Sheet. ☐
- **I agree** to take part in the above study. ☐

Project Student: Megan Brennan E-mail: mb2091@hw.ac.uk	Project Supervisor: Mike Just Email: m.just@hw.ac.uk

Name:
Signature:
Date:
Email Address:

I would like to receive information on the results of this study: Yes ☐ **No** ☐

D.16. User Study Questionnaire

Cryptography Guessing Game - User Study Questionnaire

This form is for participants taking part in the study. It should only be completed if the information sheet has been read and the consent form completed and returned to mb2091@hw.ac.uk.

Participants can withdraw from the study at any point either before, during or after. Should participants wish to withdraw, please contact either myself (mb2091@hw.ac.uk) or my project supervisor (m.just@hw.ac.uk).

* Required

Pre-game Questions

1. Please enter your ID number (ask if unsure) *

2. Have you previously learnt about classical ciphers? *

- ◯ Yes
- ◯ No

3. If yes to the previous question, how confident are you in your knowledge of classical ciphers?

	Very confident	Confident	Not so confident	Unconfident
Statement 1	◯	◯	◯	◯

During/post Game Questions

This section is to be completed either as you play through the game or after, this choice is up to the participant. Some of your tasks will be timed and these times recorded in this questionnaire, however this can be noted down either during or after completing said tasks.
Please ask here what your group is.

4. What participation group are you in? *

- ◯ A
- ◯ B
- ◯ C

5. For the shift cipher challenge (easy) how long did the challenge take you? *

6. For the shift cipher challenge (hard) how long did the challenge take you? *

7. For your 3rd challenge, please note the challenge title and the time taken to complete *

8. For your 4th challenge, please note the challenge title and the time taken to complete *

9. After checking the profile page, what score did you achieve? *

10. This question is about the websites UI, please rate the following statements based on whether you strongly disagree, disagree, are neutral, agree or strongly agree. *

	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
The website was easy to navigate	○	○	○	○	○
I was able to understand how the challenges worked without confusion	○	○	○	○	○
The interface felt intuitive to use	○	○	○	○	○

11. This question is about your learning experience, please rate the following statements based on whether you strongly disagree, disagree, are neutral, agree or strongly agree. *

	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
The cipher information pages helped me understand the ciphers	○	○	○	○	○
Completing challenges helped me feel more confident in my understanding of ciphers	○	○	○	○	○
The web game made learning about cryptography more engaging	○	○	○	○	○
I would use this tool to learn classic cryptography over reading books and websites	○	○	○	○	○

12. This question is about challenge design, please rate the following statements based on whether you strongly disagree, disagree, are neutral, agree or strongly agree. *

	Strongly disagree	Disagree	Neutral	Agree	Strongly agree
The scoring/feedback system was helpful for understanding my progress	○	○	○	○	○
The challenge mechanics (e.g. input, hints, etc) were clear	○	○	○	○	○
The difficulty of the challenges felt appropriate	○	○	○	○	○

13. This question is for open feedback regarding what you liked/didn't like about the game.

Whether you would use the tool to learn classic cryptography or not and if not why?

Whether you felt like there was any aspects of the site that were confusing or difficult to use?

If you have any suggestions for improvements?